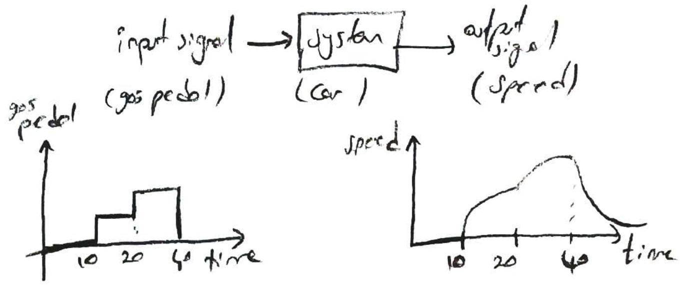
Signals and Systems

Date: 2.03.2021

- Trying to represent systems mathematically


input signal
system
output signal
(gas pedal)
(car)
(speed)
gas pedal
10
20
40
time
speed
10
20
40
time

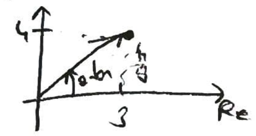
Complex Numbers:

Euler's Law = $e^{i\theta} = \cos\theta + i\sin\theta$

polar rep.   cartesian representation

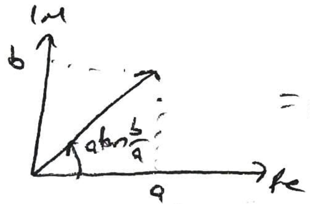
$i = \sqrt{-1}$

$3+4i \Rightarrow$


Im
4
atan 4/3
3
Re

$= \angle \text{atan}\frac{4}{3}$

$a+bi =$


Im
b
atan b/a
a
Re

$= \sqrt{a^2+b^2} \angle \text{atan}\frac{b}{a}$

** $a+ib = r\angle\theta = re^{i\theta}$ $r = \sqrt{a^2+b^2}$ $\theta = \text{atan}\frac{b}{a}$

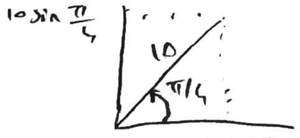
Exercise: $1+i$ write this in polar rep.

$1+i = \sqrt{2}\angle\frac{\pi}{4} = \sqrt{2}e^{i\frac{\pi}{4}}$ ✓

$3+2i = \sqrt{13}\angle\text{atan}\frac{2}{3} = \sqrt{13}e^{i\,\text{atan}\frac{2}{3}}$ ✓

$5+4i = \sqrt{41}\angle\text{atan}\frac{4}{5} = \sqrt{41}e^{i\,\text{atan}\frac{4}{5}}$ ✓

$10e^{i\frac{\pi}{4}} = 10\angle\frac{\pi}{4} = 10\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$?

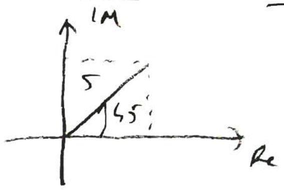

10 sin π/4
10
π/4
10 cos π/4

$3e^{-\frac{\pi}{2}?} = 3\angle\frac{-\pi}{2} = 3\left(\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2})\right) = -3i$

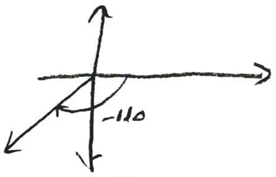
Phasor Representation:

Page 1

$5e^{i45°} \Rightarrow$



$10e^{-110° i} \Rightarrow$



$e^{i\frac{\pi}{2}} = i$ $e^{i\frac{3\pi}{2}} = -i$ $e^{i\pi} = -1$ $e^{i2\pi} = 1$

$e^{-i2\pi} = 1$ $e^{-i\frac{3\pi}{2}} = i$

$\propto$ $e^{i101\pi} = e^{i2\pi.50 + i\pi} = e^{i2\pi50} . e^{i\pi} = 1 . -1 = -1$ ✓

$\propto$ $e^{127\pi} = e^{i2\pi.64} . e^{-i\pi} = 1 . -1 = -1$ ✓

$\propto$ $e^{\frac{127}{3}\pi} = e^{i(42+\frac{1}{3})\pi} = e^{i42\pi} . e^{i\frac{\pi}{3}} = e^{i\frac{\pi}{3}}$

$e^{i\frac{\pi}{3}} = e^{i60°} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$

Magnitude and angle of a complex

$a+ib = re^{i\theta}$ → angle   ↑ magnitude $|a+ib| = \sqrt{a^2+b^2}$

Complex Conjugate in Polar Representation

$(a+ib)^* = \overline{a+ib} = a-ib$

$(re^{i\theta})^* = re^{-i\theta}$

$\overline{r(\cos\theta + i\sin(\theta))} = r(\cos\theta - i\sin\theta)$

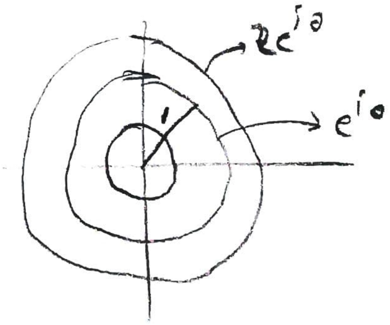
$|e^{i\theta}| = \sqrt{e^{i\theta} . e^{-i\theta}} = 1$

$|z| = \sqrt{z . z^*}$ $z = re^{i\theta}$

Locus of $e^{i\theta}$

$z = e^{i\theta}$

unit circle


1
$2e^{i\theta}$
$e^{i\theta}$

Question: $\cos(i)$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{-i\theta} = \cos(-\theta) + i\sin(-\theta) = \cos\theta - i\sin\theta$$

+

$$e^{i\theta} + e^{-i\theta} = 2\cos\theta \Rightarrow \cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}$$

$$\cos(i) = \frac{e^{ii}+e^{-ii}}{2} = \frac{e^{-1}+e}{2} = \frac{\frac{1}{e}+e}{2}$$

$$\Rightarrow \frac{\frac{1}{2.71}+2.71}{2} = \ldots$$

Question:

$a\cos(2)$

$$2 = \frac{e^{i\theta}+e^{-i\theta}}{2} \qquad e^{i\theta} = x$$

$$4 = e^{i\theta} + e^{-i\theta} \qquad x^2 - 4x + 1 = 0$$

$$4 = x + \frac{1}{x} \qquad x = 2 \pm \sqrt{3}$$

$$e^{i\theta} = 2 \pm \sqrt{3}$$

$$i\theta = \ln(2 \pm \sqrt{3})$$

$$\boxed{\theta = i\ln(2 \mp \sqrt{3})}$$

→ bunu böyle yazdı

bence $\Rightarrow \theta = \frac{\ln(2 \mp \sqrt{3})}{i}$

Date: 4.03.2021

Phasor Representation

$$z = x + iy = Re^{i\theta}$$

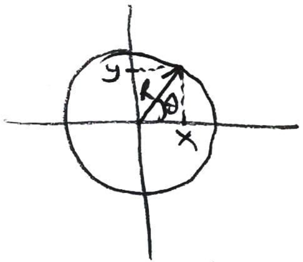
$$R = |z| = \sqrt{x^2+y^2}$$

$$\theta = \angle z = \operatorname{atan}\frac{y}{x}$$

$$R\,e^{i\overset{\theta}{\widetilde{\omega t}}}$$

angular frequency

time



1 rotation $\frac{2\pi}{\omega}$ secs

$$T = \frac{2\pi}{\omega} \quad f = \frac{1}{T} = \frac{\omega}{2\pi}$$

$$\omega = 2\pi f$$

$$e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, dt$$ (Fourier Transforms)

Example:

$$\ln(-10) = \ln\left(10\, e^{i\pi + 2k\pi i}\right) =$$

$$\ln 10 + \ln e^{2k\pi i + i\pi}$$

$$= \ln 10 + i\pi + 2k\pi i \quad ✓$$

Example:

$$\ln(10) = \ln\left(10\, e^{2\pi k i}\right) = \ln 10 + 2\pi k i \quad ✓$$

Example (square root)

$$\sqrt{4} = \left(4e^{i2\pi k}\right)^{1/2} = 2e^{i\pi k} \begin{cases} \rightarrow 2 & k = 0, 2, 4, -2, \ldots \\ \rightarrow -2 & k = 1, 3, -1, -3 \end{cases}$$

$$\sqrt{-4} = \left(4e^{i\pi + 2i\pi k}\right)^{1/2} = 2e^{i\frac{\pi}{2} + i\pi k} \begin{cases} \rightarrow 2e^{i\frac{\pi}{2}} = 2i \\ \rightarrow 2e^{-i\frac{\pi}{2}} = -2i \end{cases}$$

* kök derecesi kaç ise o kadar solution olur

Signals

continuous

- $e^{\alpha t}$
- $\sin(\omega t)$, $\cos(\omega t)$
- $\chi_{(a,b)}(t)$
- $N(t)$
- $\delta(t)$
- $\operatorname{sinc}(t)$

functions

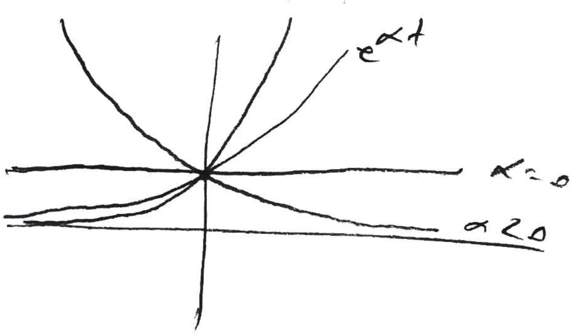
Discrete

- $e^{\alpha n}$
- $\sin(\Omega n)$, $\cos(\Omega n)$
- $\chi_{(a,b)}(n)$

sequences

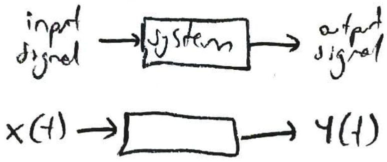
Exponent

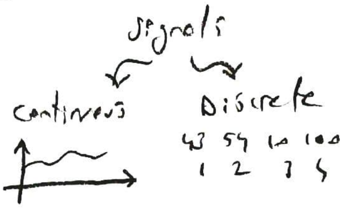
$e^{\alpha t}$



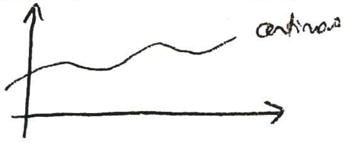
Signals and Systems

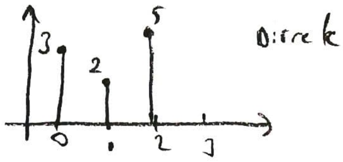
Date: [illegible].03.2021


input signal
system
output signal
x(t)
y(t)

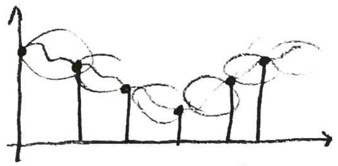

Signals
continuous
Discrete
43 54 10 100
1 2 3 4

continuous signals





Temperature Example:



> Bilgisayar bunu böyle görmez

- exponential signals
- sinusoid signals

$A\sin(\omega t+\alpha), A\cos(\omega t+\alpha)$

angular frequency phase amplitude

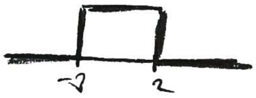
$f = \frac{\omega}{2\pi}$ hertz (1/sec) $\omega = 2\pi f$

$T = \frac{1}{f} = \frac{2\pi}{w}$

Indicator Function:

$$\chi_{(a,b)}(t) = \begin{cases} 1 & a<t<b \\ 0 & \text{else} \end{cases}$$

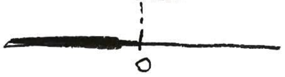
Exercise: Draw $\chi_{(-3,2)}(t)$



Unit Step:

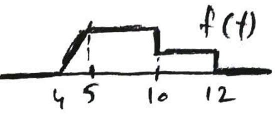
$$U(t) = \begin{cases} 1 & t>0 \\ 0 & t<0 \end{cases}$$

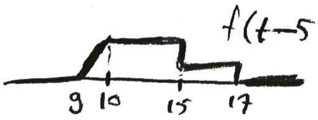
(t=0) belirsiz

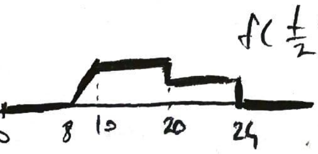


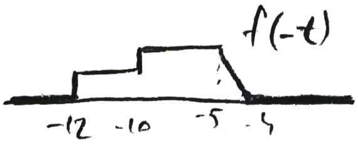
Transformation of Signals:

$$f(t) = \begin{cases} f(t-a) & \text{translated by } a \\ f(at) & \text{scaled by } a \end{cases}$$





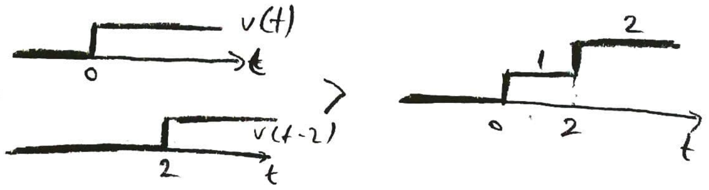




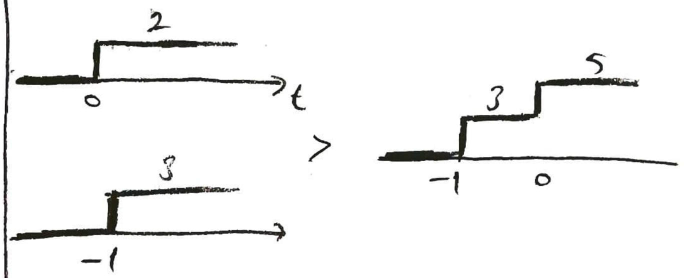
Adding Two Signals

Date: 11.03.2021

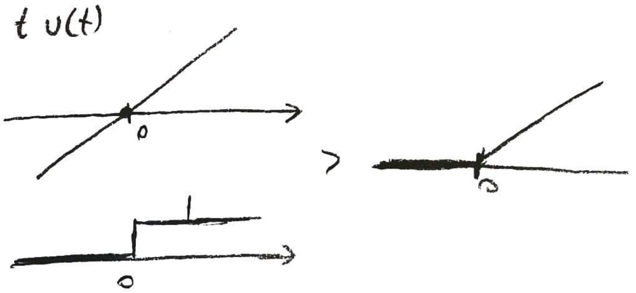
* $u(t) + u(t-2)$



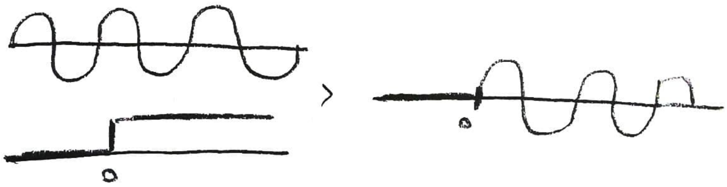
* $2u(t) + 3u(t+1)$



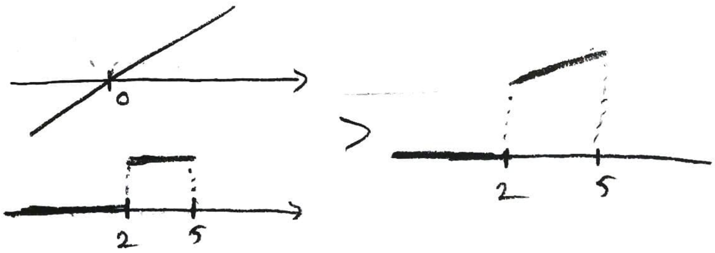
* $t\,u(t)$



* $\sin(t)\,u(t)$

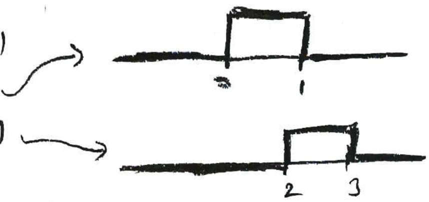


* $t\,\chi_{(2,5)}(t)$

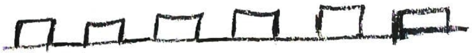


* $\sum_{i=-\infty}^{i=\infty} \chi_{(0,1)}(t-2i)$

$= \dots + \chi_{(0,1)}(t+4)$
$+ \chi_{(0,1)}(t+2)$
$+ \chi_{(0,1)}(t)$
$+ \chi_{(0,1)}(t-2)$
$+ \chi_{(0,1)}(t-4)$

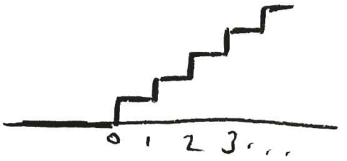


$\Rightarrow$



* $\sum (t-2n)\chi_{(0,1)}(t-2n)$ (left for me) (sawtooth)

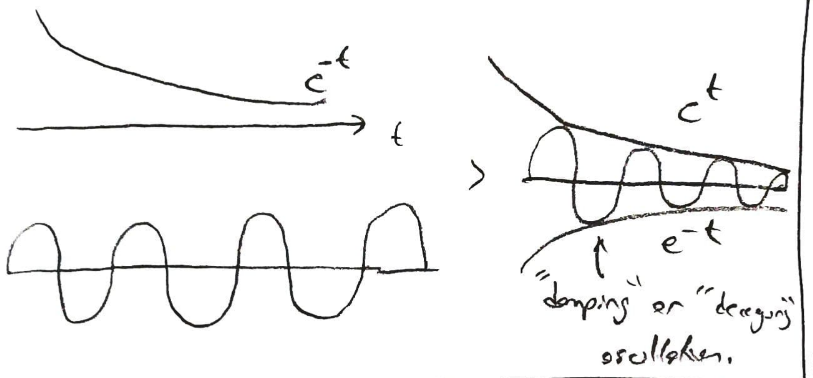
* $\sum_{r=0}^{\infty} N(t-r)$



"stairway to heaven"

Multiplication of sinusoid with Exponential

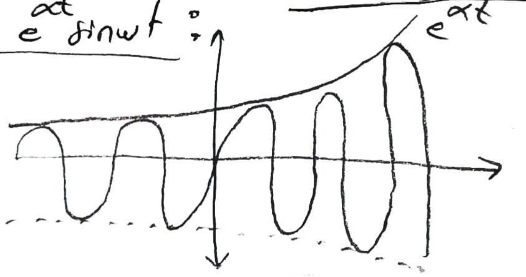
* plot $\sin(t)\, e^{-t}$



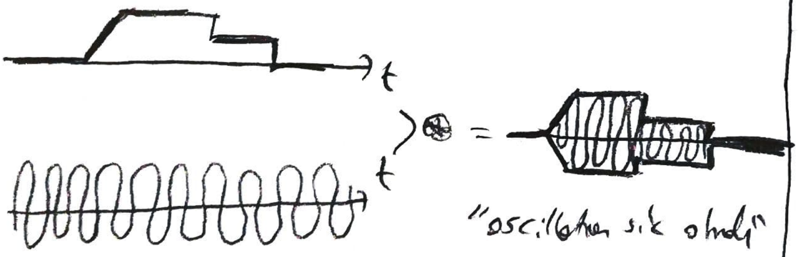
"damping" or "decaying" oscillation.

Date: 16.03.2021

Multiplication of $e^{\alpha t}\sin\omega t$:



Multiplication of f(t) by $\sin\omega t$



"oscillation sik ohndi"

Function norms (signal norms)

Norm of a vector = length of a vector = magnitude of a vector

* $L2(f(t)) = \sqrt{\int_{-\infty}^{\infty} f(t)^2 dt}$

$L^2\left(\begin{bmatrix} +1 \\ -2 \\ 5 \end{bmatrix}\right) = \sqrt{1^2+(-2)^2+5^2} = \sqrt{30}$

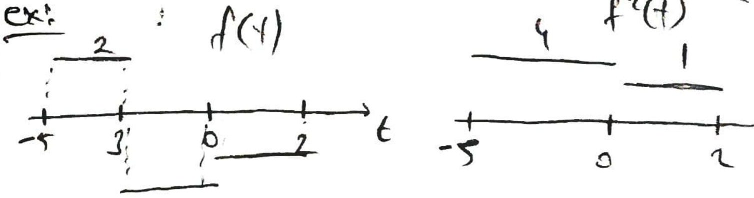
$L2\left(\begin{bmatrix} a_1 \\ a_2 \\ a_k \end{bmatrix}\right) = \sqrt{\sum_{i=1}^{k} |a_i|^2}$

* $L1(f(t)) = \int_{-\infty}^{\infty} |f(t)|\, dt$

$L1\left(\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}\right) = |1| + |-2| + |5| = 8$

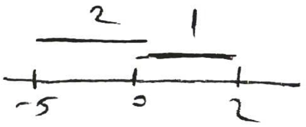
$L1\left(\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_k \end{bmatrix}\right) = \int_{-\infty}^{\infty} |f(t)|\, dt$

ex:



find $\|f(t)\|_{L2} = \sqrt{\int_{-\infty}^{\infty} |f(t)|^2 dt} = \sqrt{22}$

$\|f(t)\|_{L1} = 12$



$\|e^{t}\|_{L2} = \sqrt{\int_0^{\infty} (e^{t})^2 dt} = \sqrt{\int_0^{\infty} e^{2t} dt} = \infty$

$\|e^{-2t}u(t)\|_{L2} = \sqrt{\int_{-\infty}^{\infty} (e^{-2t}u(t))^2 dt} = \sqrt{\int_{-\infty}^{\infty} e^{-4t}u(t)\, dt}$

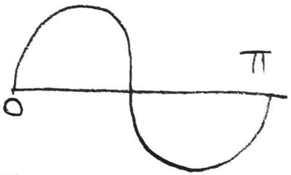
$\|e^{-2t}u(t)\|_{L1} = \sqrt{\int_0^{\infty} e^{-4t} dt} = 1/2$

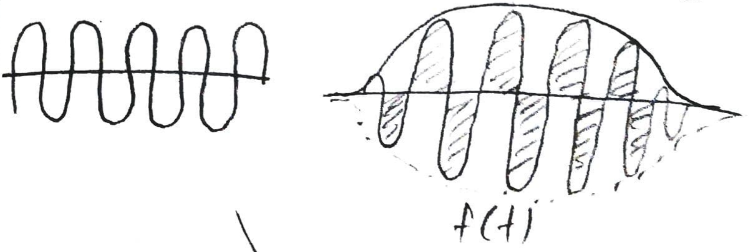
??

Riemann-Lebesgue Theorem

Let us have f(t) with finite L^1norm, i.e, $\|f(t)\|_{L1} < \infty$. Then

$$\lim_{\omega\to\infty} \int_{-\infty}^{\infty} f(t)\sin\omega t\, dt = 0$$





$$\cancel{\lim_{\omega\to 0} \int_{-\infty}^{\infty} e^{t}\sin\omega t\, dt \longrightarrow 0}$$

Dirac Deltas

A signal which has area 1 over point 0 and area 0 in every interval that does not contain point 0

Date: 18.03.2021

Sinc function

- $\mathrm{sinc}(t) = \frac{\sin(t)}{t} = \frac{1}{t} \cdot \sin(t)$
- $a \cdot \frac{\sin(at)}{at} = \frac{\sin(at)}{t}$

"grafik vardı atladım"

$$f(0) = \lim_{t \to 0} \frac{\sin(at)}{t} = \lim_{t \to 0} \frac{a\cos(at)}{1} = a$$

Date: 23.03.2021

"grafik var bakarsın"

Area under the sinc(x) curve

$$\int_{-\infty}^{\infty} \mathrm{sinc}(x)\,dx = \int_{-\infty}^{\infty} \frac{\sin(x)}{x}\,dx = \pi$$

"Just memorize it"

* dirichlet integral

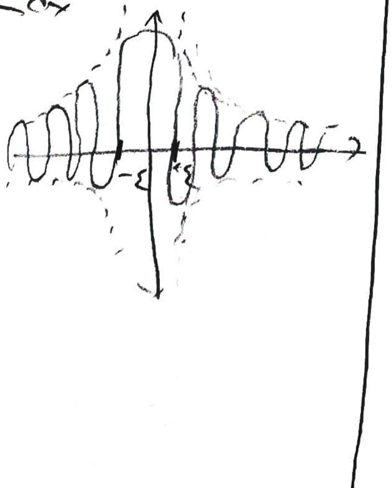
- $\int_{0}^{\infty} \mathrm{sinc}(x)\,dx = \frac{\pi}{2}$
- $\int_{-\infty}^{\infty} a\,\mathrm{sinc}(ax)\,dx = \int_{-\infty}^{\infty} \frac{\sin(ax)}{ax} \cdot a\,dx = \int_{-\infty}^{\infty} \frac{\sin(ax)}{x}\,dx$

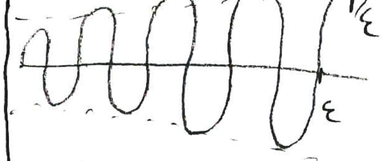
$ax = u \quad a\,dx = du \quad dx = \frac{du}{a} \quad x = u/a$

$$\boxed{\int_{-\infty}^{\infty} \frac{\sin(u)}{u}\,du = \pi}$$

$$\int_{-\infty}^{\infty} a\,\mathrm{sinc}(ax)\,dx = \int_{-\infty}^{\infty} \frac{\sin(ax)}{x}\,dx$$

$$= \int_{-\infty}^{-\varepsilon} \frac{\sin(ax)}{x}\,dx + \int_{-\varepsilon}^{+\varepsilon} \frac{\sin(ax)}{x}\,dx + \int_{+\varepsilon}^{\infty} \frac{\sin(ax)}{x}\,dx = \pi$$





"ε area is 0 because of R-L

Actually it is not exactly because of R-L, because R-L says:

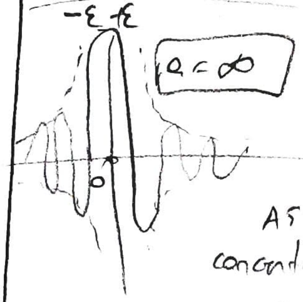
$$\lim_{a \to \infty} \int_{-\infty}^{\infty} f(x) \sin ax\,dx = 0$$ and $f(x)$ must be L1

and here our $f(x) = \frac{1}{x} u(t-\varepsilon)$ this is not L1

So, R-L does not work directly. But there is a small trick which makes it work.

Definition of L1: A function $f(x)$ is L1 if $\int_{-\infty}^{\infty} |f(x)|\,dx < \infty$. L1 norms of $f(x)$ is:

$\|f(x)\|_{L^1} = \int_{-\infty}^{\infty} |f(x)|\,dx$, L2 norm is $\|f(x)\|_{L^2} = \sqrt{\int_{-\infty}^{\infty} |f(x)|^2\,dx}$



$$\lim_{a \to \infty} \int_{x}^{y} \frac{\sin(ax)}{x}\,dx = \begin{cases} 0, & 0 \notin (x,y) \\ \pi, & 0 \in (x,y) \end{cases}$$

As $a \to \infty$, all the area of $\frac{\sin(ax)}{x}$ is concentrated on top of the point $x=0$

$\frac{\sin(ax)}{x} \quad a = 1 \quad \frac{\sin(a\pi/4)}{\pi/4} \begin{cases} a=1 & \frac{\sqrt{2}/2}{\pi/4} \\ a=2 & 4/\pi \\ a=\infty & ? \end{cases}$

Date: 25.03.2021

- Let $f(t) \in L^1$

$$\lim_{t \to \infty} \int_{-\infty}^{\infty} f(t) \sin wt\,dt = 0$$

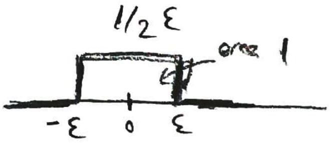
Every physical measurement is spread over some small time, hence every measurement is an area measurement

"Infinitely fast oscillations can be taken as zero"

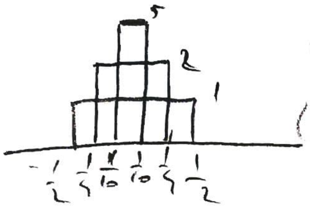
Page 5

Dirac Delta

Has an area 1 on the single point 0, has area zero on any interval not including point 0.



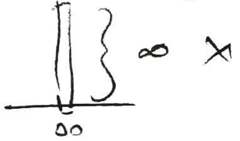
$$\frac{1}{2\varepsilon}\chi_{(-\varepsilon,\varepsilon)}(t)$$



$$\boxed{\lim_{\varepsilon\to0}\frac{1}{2\varepsilon}\chi_{(-\varepsilon,\varepsilon)}(t)}$$

area = 1

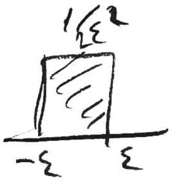
* Can we say that dirac delta is 0 in ∞



NO!

$$\chi_{(a,b)}(t) = \begin{cases}1 & a<t<b\\ 0 & \text{else}\end{cases}$$

* Let us consider $\lim_{\varepsilon\to0}\frac{1}{2\varepsilon^2}\chi_{(-\varepsilon,\varepsilon)}(t)$

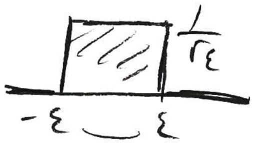


area = 1/ε

puts infinite area on top of point 0 ⇒ Not dirac delta

* Lets us consider $\lim_{\varepsilon\to0}\frac{1}{\sqrt{\varepsilon}}\chi_{(-\varepsilon,\varepsilon)}(t)$

puts 0 area on top of point ⇒ not a dirac delta



Area = 2√ε → 0

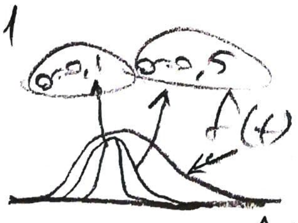
* How to construct a dirac delta?

Take a function f(t) with area 1 under it

$$\int_{-\infty}^{\infty} f(t)\,dt = 1$$

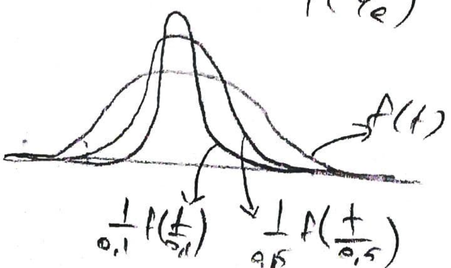
consider

$$\int \frac{1}{a} f\left(\frac{t}{a}\right)dt$$



$f(t/\varepsilon)$

$$\int_{-\infty}^{\infty}\frac{1}{a} f\left(\frac{t}{a}\right)dt = \int_{-\infty}^{\infty} f(u)\,du$$



f(t)
$\frac{1}{0,1}f(\frac{t}{0,1})$
$\frac{1}{0,5}f(\frac{t}{0,5})$

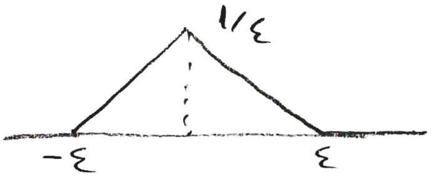
Area = 1 all cases

* $\lim_{a\to0}\frac{1}{a} f\left(\frac{t}{a}\right) = \delta(t)$

Page = 6

Date: 30.03.2021

Ex:



$$\text{Area} = \frac{1}{\varepsilon}\cdot 2\varepsilon\cdot\frac{1}{2} = 1$$ ✓

ex:

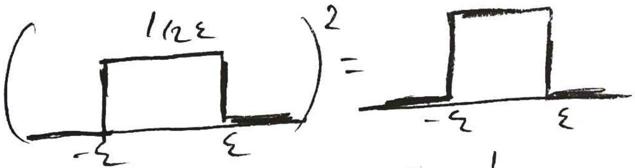
$$\frac{1}{\sqrt{2\pi}}e^{-\frac{t^2}{2}}$$

$$\lim_{\varepsilon\to0}\frac{1}{\sqrt{2\pi}\,\varepsilon}e^{-\frac{1}{2}\left(\frac{t}{\varepsilon}\right)^2} = \delta(t)$$

Properties of δ(t):

1) Some Math operations of δ(t) is prohibited.

Consider $[\delta(t)]^2$



becomes δ(t) as ε→0

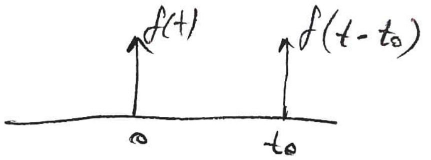
$\text{area} = \frac{1}{2\varepsilon}$

2) Multiplying with a constant

$5\delta(t)$ $3\delta(t)$ $\pi\delta(t)$ $-2\delta(t)$

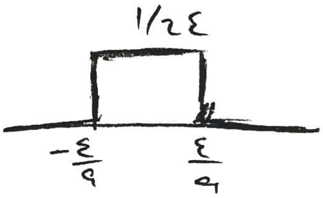
3) Translation

$\delta(t-t_0)$



4) Scaling

$$\lim_{\varepsilon\to0}\frac{1}{2\varepsilon}\chi_{(-\varepsilon,\varepsilon)}(at)$$



$$\boxed{\delta(at) = = \frac{1}{|a|}\delta(t)}$$

5) Multiplication by a function

$f(t)\in L^1$

important

$$\boxed{f(t)\cdot\delta(t) = f(0)\cdot\delta(t)}$$

Ex: $\underbrace{\chi_{(-1000,1000)}(t)\,t^2\,\delta(t-4)}_{\text{lives only at } t=4} = 16\,\delta(t-4)$

$$f(t)\,\delta(t-t_0) = f(t_0)\,\delta(t-t_0)$$

Ex: $\chi_{(-1000,1000)}(t)\,[t^2-2t]\,[\delta(t-1200)+2\delta(t+4)-\delta(t-2)]$

$t=1200 \to$ out of region so it is $=0$
$t=-4 \to 24$
$t=2 \to 0$

$$\text{comp}= 48\,\delta(t+4)$$

Ex: $t^2\chi_{(-1000,1000)}(t)\;\left(\frac{2}{3}\delta(t+2)-\frac{1}{3}\delta(t-2)\right)$

$t^2 \to L^1$ değil
o yüzden
diminligence
$(-1000,1000)$ diye

lives at
$t=-2 \to 4$
$t=2 \to 4$

$$\Rightarrow \frac{8}{3}\delta(t+2)-\frac{4}{3}\delta(t-2)$$

Quiz Sorusu : find the L^2 norm of $\frac{u(t-2)}{t}$

$$\sqrt{\int_{-\infty}^{\infty}\left|\frac{u(t-2)}{t}\right|^2 dt}=\sqrt{\int_{-\infty}^{2}\frac{0}{t^2}dt+\int_{2}^{\infty}\frac{1}{t^2}dt}$$

$$=\sqrt{\int_2^{\infty}\frac{1}{t^2}dt}=\sqrt{-\frac{1}{t}\Big|_2^{\infty}}=\frac{1}{2}\,//$$

(Özelliklere devam)

6) $$\int_{-\infty}^{\infty}f(t)\delta(t)\,dt=\int_{-\infty}^{\infty}f(0)\delta(t)\,dt=f(0)\int_{-\infty}^{\infty}\delta(t)\,dt=f(0)$$

$$\boxed{\int_{-\infty}^{\infty}f(t)\delta(t)\,dt=f(0)}$$

for example:

$$\int_{-\infty}^{\infty}\underbrace{\frac{\cos(2t)e^t}{1+e^{t^2}+t^4}}_{\text{lives at }=0}\delta(t)\,dt=\frac{1}{2}\,//$$

Dirac delta kills integrals

Ex: $\int_{-\infty}^{\infty}\frac{\cos(2t)e^t}{1+e^{t^2}+t^4}\left[\frac{1}{2}\delta(t-1)-\delta(t-2)\right]dt$

$t=1 \quad t=2$

$$\frac{1}{2}\frac{(\cos 2)e}{1+e+1}\delta(t-1)-\frac{\cos(4)e^2}{1+e^4+2^4}\delta(t-2)$$

* Can sinc function be made Dirac delta?

$$f(t)\to\lim_{\varepsilon\to0}\frac{1}{\varepsilon}f\left(\frac{t}{\varepsilon}\right)$$

$$\frac{1}{\pi}\text{sinc}(t)\to\lim_{\varepsilon\to0}\frac{1}{\pi}\cdot\frac{1}{\varepsilon}\text{sinc}\left(\frac{t}{\varepsilon}\right)$$

$$=\lim_{\varepsilon\to0}\frac{1}{\varepsilon}\frac{\sin t/\varepsilon}{t/\varepsilon}=\lim_{\varepsilon\to0}\frac{\sin\left(\frac{t}{\varepsilon}\right)}{t}$$

$$\int f(t)\lim_{\varepsilon\to0}\underbrace{\frac{1}{\varepsilon\pi}\text{sinc}\left(\frac{t}{\varepsilon}\right)}_{\text{if this is }\delta(t)}dt=f(0)$$

Date: 01.04.2021

$$\delta(t)=\lim_{\varepsilon\to0}\frac{1}{\varepsilon}\frac{1}{\pi}\cdot\text{sinc}\left(\frac{t}{\varepsilon}\right)=\lim_{\varepsilon\to0}\frac{1}{\pi\varepsilon}\frac{\sin\left(\frac{t}{\varepsilon}\right)}{t/\varepsilon}$$

$$=\lim_{\varepsilon\to0}\frac{1}{\pi t}\sin\left(\frac{t}{\varepsilon}\right)\ \overset{\text{equal}}{\Leftrightarrow}\ \lim_{a\to0}\frac{\sin(at)}{t}$$

$$\to\int_{-\infty}^{\infty}f(t)\lim_{\varepsilon\to0}\frac{1}{\varepsilon t}\sin\left(\frac{t}{\varepsilon}\right)dt$$

$$=\cancel{\int_{-\infty}^{-\delta}f(t)\frac{1}{\varepsilon t}\sin\left(\frac{t}{\varepsilon}\right)dt}\to0+\int_{-\delta}^{\delta}\frac{f(t)}{\varepsilon t}\sin\left(\frac{t}{\varepsilon}\right)dt+\cancel{\int_{\delta}^{\infty}\frac{f(t)}{\varepsilon t}\sin\left(\frac{t}{\varepsilon}\right)dt}\to0$$

$$=\int_{-\delta}^{\delta}\frac{f(t)}{\varepsilon t}\sin\left(\frac{t}{\varepsilon}\right)dt=f(0)\int_{-\delta}^{\delta}\frac{1}{\varepsilon t}\sin\left(\frac{t}{\varepsilon}\right)dt=f(0)$$

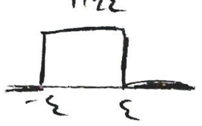
(δ işareti dirac delta değil burada)

$$\Rightarrow\boxed{\frac{d}{dt}u(t)=\delta(t)}$$ olay bu.

$$f'(t)=\frac{d}{dt}f(t)=\lim_{\varepsilon\to0}\frac{f(t+\varepsilon)-f(t-\varepsilon)}{2\varepsilon}$$

$$\frac{d}{dt}u(t)=\lim_{\varepsilon\to0}\frac{u(t+\varepsilon)-u(t-\varepsilon)}{2\varepsilon}$$

$$\lim_{\varepsilon\to0}\frac{\chi_{(-\varepsilon,\varepsilon)}(t)}{2\varepsilon}=\delta(t)$$



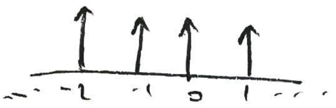
proved: $\frac{d}{dt}u(t)=\delta(t)$

prove: $\int_{-\infty}^{x}\delta(s)\,ds=\underbrace{\begin{cases}0, & x<0\\1, & x>0\end{cases}}_{u(x)}$ dont consider $x=0$

Dirac Delta Train

= impulse train

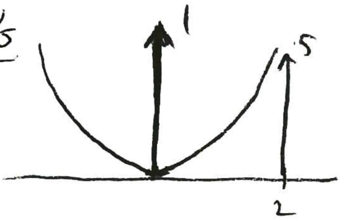
$$\sum_{n=-\infty}^{\infty} \delta(t-n)$$



dirichlet's comb

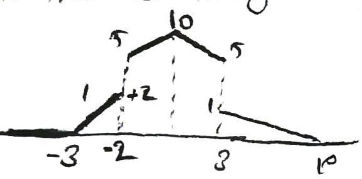
Plotting functions with $\delta(t)$'s

$$t^2 + \delta(t) + 5\delta(t-2)$$

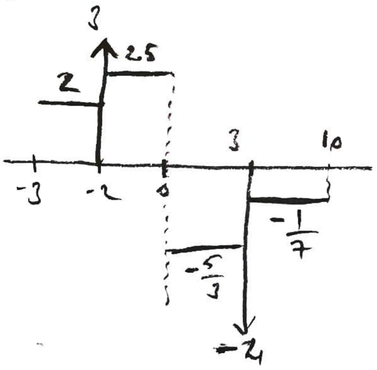


Derivative of functions

whenever there is a discontinuity, in the derivative there is a dirac delta = impulse equal in size is the discontinuity.



10
5
5
1
+2
1
-3
-2
3
10

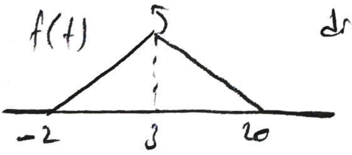


3
2.5
2
3
10
-3
-2
0
$-\frac{1}{7}$
$-\frac{5}{3}$
-4

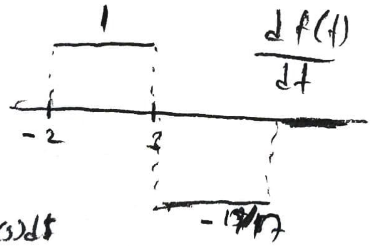
when integrating, whenever we encounter a dirac delta we get a discontinuity.

Date: 06.04.2021

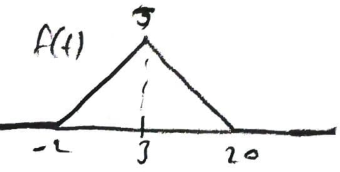
Differentiation and Integration of Continuous Signals



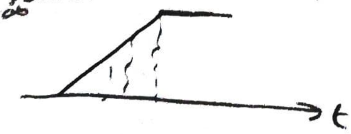
diff
=>



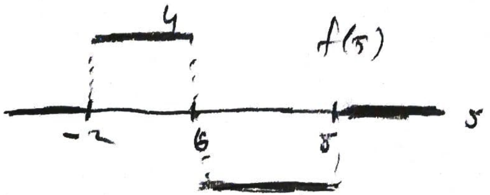
1
$\frac{df(t)}{dt}$
-2
3
-1/17

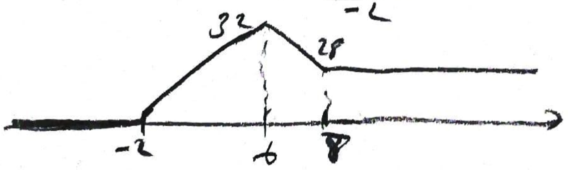


$$\int_{-\infty}^{t} f(s)\,ds$$

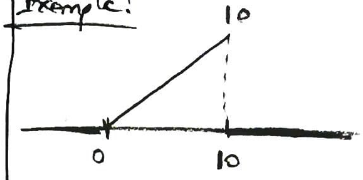


Ex:

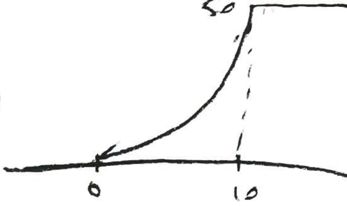




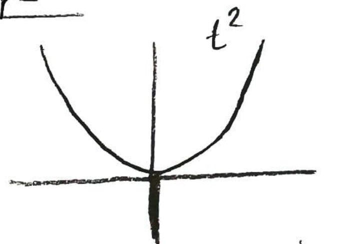
Example:



=>



Example



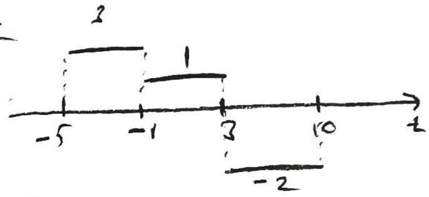
$$\int_0^t s^2\,ds$$

*our main problem is when we have dirac deltas

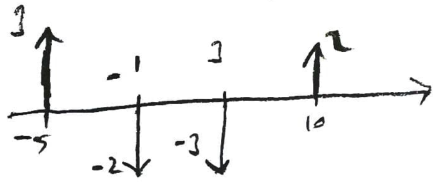
$$\int_{-\infty}^{t} \delta(s)\,ds = u(t)$$

- integral of dirac delta is discontinuity.
- derivative of discontinuity is dirac delta

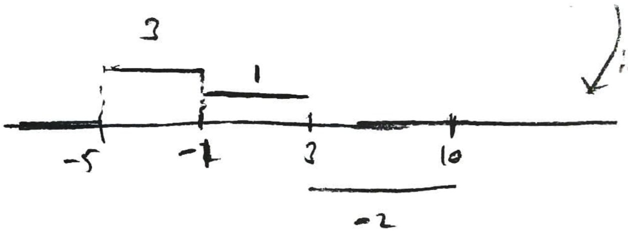
Ex:



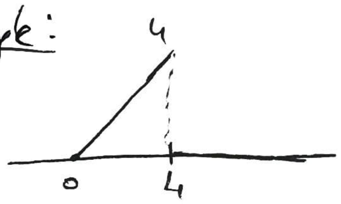
derivative



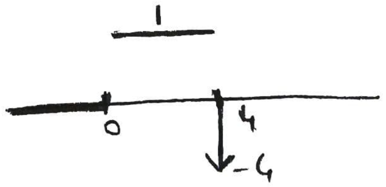
integral



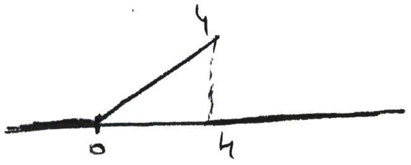
Example:



$\frac{d}{dt}$ ↓

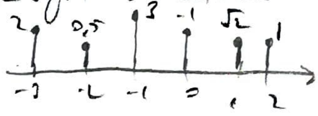


$\int_{-\infty}^{t} \dots$ ↓



Discrete Signals

→ Discrete signals are defined at integer values.



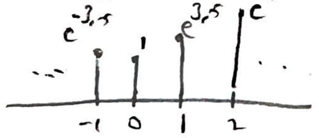
continous signals are denoted by $x(t)$. Discrete signals are denoted by $x[n]$. 3 nokta yoksa onlar sıfırdır.

$0\,0\,0\,0\,0\,0 \quad 2 \quad 5 \quad -1 \quad \underline{3} \quad \sqrt{2} \quad -4.1 \quad 1 \quad 0\,0\,0\,0\,0\,0$

$\dots \; 3 \; \underline{2} \; 5 \; \dots \neq 0\,0\,0\,0\,0 \; 3 \; \underline{2} \; 5 \; 0\,0\,0\,0$

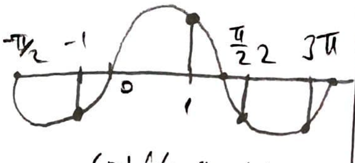
exponentials

$x[n] = e^{3.5n}$



sinusoids

$x[n] = \sin(2n)$

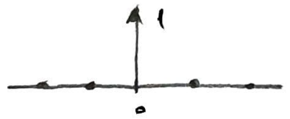


(noktalar sinyal)

$\sin(\pi t)$ is not 0

$\sin[\pi n] = 0$

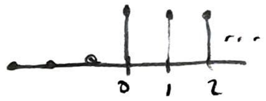
Dirac Delta :



$\delta[n]$

$\underline{1}$

unit step:



$u[n]$

shifting: $f[\underbrace{n-n_0}]$, must be integer.

scaling: not defined.

sinusoids (Again) :

If you have a periodic signal like $\sin(\omega t)$, then discrete version $\sin[\Omega n]$ may not be periodic.

Date: 8.04.2021

$f(t)$
↑
continous

$f[n]$
↑
discrete

e^{t}

e^{n}

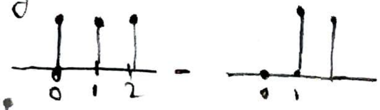
- $e^{\alpha n}$
- $\sin[\Omega n]$, $\cos[\Omega n]$

- $u[n] = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$
- $\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$

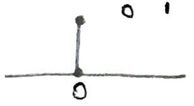
* Discrete version of dirac delta is known as kronecker delta.

- summing, multiplying, subtracting etc of discrete signals work the same way

$u[n] - u[n-1] =$



$=$



$\rightarrow \delta[n]$

Discrete sinusoid signals :

$\sin(\omega t)$	$\sin[\Omega n]$
- periodic for all w - period is $T = \frac{2\pi}{w}$ - if w increases, period decreases	- Not periodic for all Ω. Ω must be of the form $\frac{2\pi k}{N} = \Omega$ - Ω and $\Omega + 2k\pi$ has same period

periodicity

$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$

rather than investigating the periodicity of $\cos[\Omega n]$ we investigate the periodicity of $e^{j\Omega n}$. Assume the period of $e^{j\Omega n}$ is N. N must be an integer.

$f[n+N] = f[n]$ for all N. N must be an integer

↳ periodun tanımı

$e^{j\Omega(n+N)} = e^{j\Omega n}$

$\cancel{e^{j\Omega n}} \cdot e^{j\Omega N} = \cancel{e^{j\Omega n}}$

$e^{j\Omega N} = 1$

$\Omega N = 2\pi . k$

$$\boxed{\Omega = \frac{2\pi k}{N}}$$

$\cos\left[\frac{2\pi}{7} n\right]$, periodic $= 7$

$\sin[\sqrt{3}\, n]$ not periodic because we can not find two integer such that $\frac{2\pi k}{N} = \sqrt{3}$

$\sin(\sqrt{3}\, t)$, periodic $\Rightarrow T = \frac{2\pi}{\sqrt{3}}$

$\sin[2n]$ $\quad \frac{2\pi k}{N} = 2$, $\pi = \frac{N}{k}$ it is not possible.

Ex: Find the period of $\cos\left[\frac{\pi}{7} n\right]$

$\cos\left[\frac{2\pi}{14} n\right]$. $\quad N = 14$

α Find the period of $\cos\left[\frac{2\pi}{7} n\right]$ $\quad \boxed{N = 7}$

α " " " " $\cos\left[\frac{3\pi}{7} n\right]$

$\cos\left[\frac{2\pi}{14} \cdot ③ n\right]$ $\; k \quad N = 14$

If we increase Ω, will the period of N decrease?
In continuous signal $\cos(\omega t)$ as $\omega \nearrow \infty$, $T \searrow 0$
In $\cos[\Omega n]$, if $\Omega \nearrow \infty$ will $N \searrow 0$. The smallest N is $N=2$

Simplify $e^{\frac{j118\pi}{3}n} = e^{j2\pi \cdot \frac{59}{3}n}$

$$= e^{j(2\pi(19+\frac{2}{3}))n} = \cancel{e^{j2\pi 19 n}} \cdot e^{j2\pi\frac{2}{3}n}$$

as n is integer (cancels)

$$= e^{j\frac{4\pi}{3}n} \text{ with } \boxed{N=3}$$

Find equivalent to $e^{j\frac{2\pi}{3}n} = e^{j(\frac{2\pi}{3}+2\pi)n}$

$$= e^{j\frac{8\pi}{3}n}$$

$$= e^{j(\frac{2\pi}{3}+20\pi)n}$$

Jilet firef bunları fu not gret liyor.

In continuous case $e^{j\frac{2\pi}{3}t}$, $e^{j\frac{8\pi}{3}t}$, $e^{j\frac{42\pi}{3}t}$ are all different

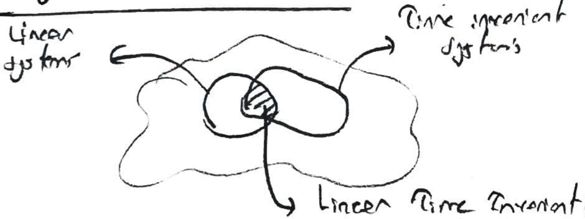
In discrete case $e^{j\frac{2\pi}{3}n}$, $e^{j\frac{8\pi}{3}n}$, $e^{j\frac{42\pi}{3}n}$ are all same.

Date: 13-04-2021

SYSTEMS

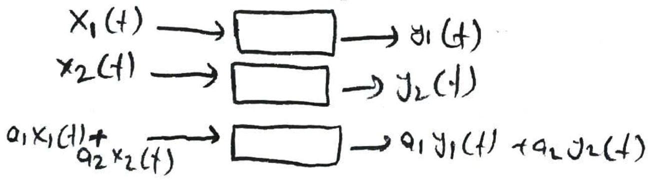
A system has an input and output port. It receives a signal from its input port and generate a response from its output port.

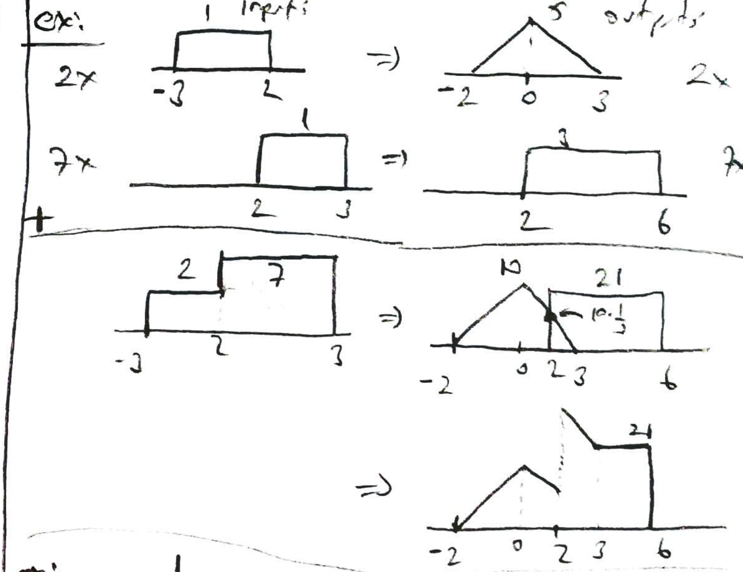
Systems in General

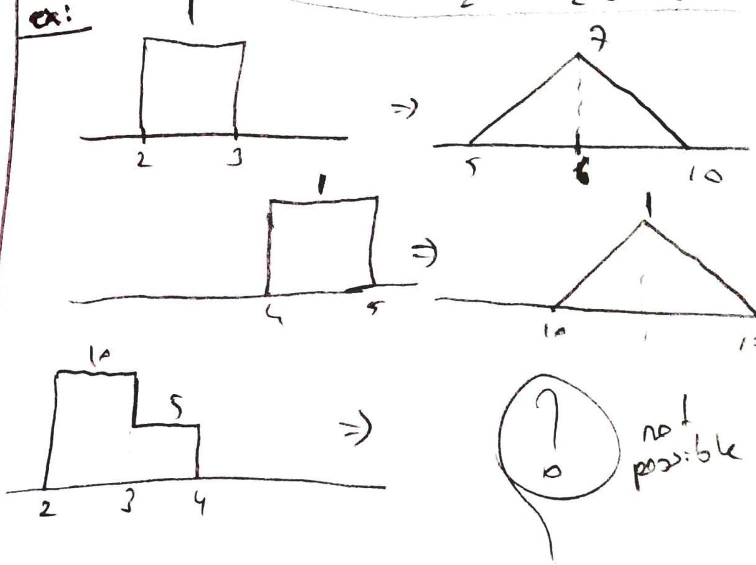


Linear Time Invariant (LTI) systems can be solved with convolution.

Linearity

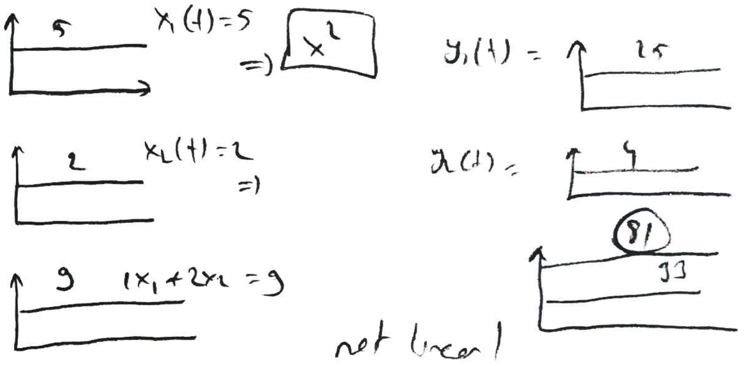






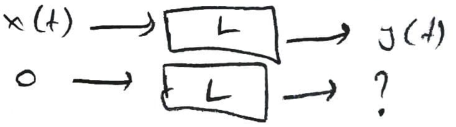
Linear Combination: If we have 2 signals $x(t)$, $y(t)$ their linear combination is $a\,x(t) + b\,y(t)$

non-linear example:

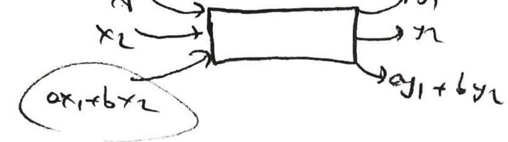


not linear!

→ If I enter 0 to a linear system, what happens?

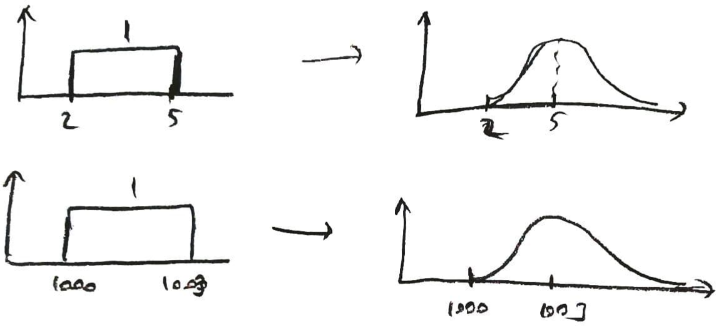


of course 0



a ve b'yi değişerek sonsuz lineer comb. üretilebilir bu özellik superposition olarak bilinir. Aksi tek örnek, olmadığını ispatlamaya yeter.

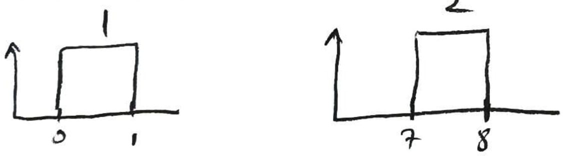
Time Invariance

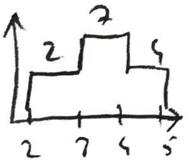


Time invariance:
$$x(t) \longrightarrow y(t)$$
$$x(t-\Delta) \rightarrow y(t-\Delta)$$

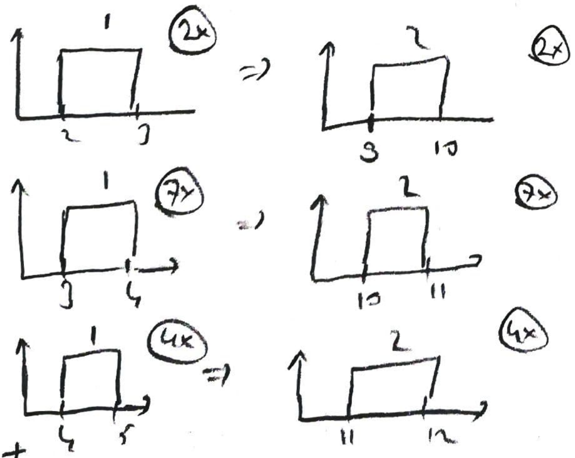
Time invariant systems does not change in time.
↳ short times için doğru.

ex: LTI system:

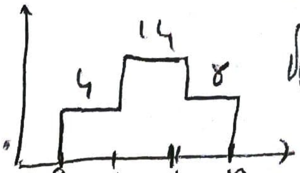




?



t



$\boxed{\text{Answer}}$

$x(t) \longrightarrow \boxed{\text{Rule}} \rightarrow y(t)$ Cont. system

$x[n] \rightarrow \quad\quad \rightarrow y[n]$ disrek system.

Date: 15.06.2021

- Given the input and the system properties (impulse response) we want to compute the output

$x(t)$ + system properties ⇒ $y(t)$

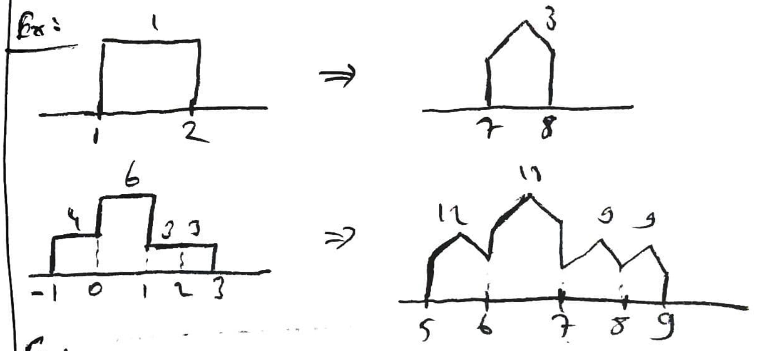
Properties of systems

- LTI systems are solvable.

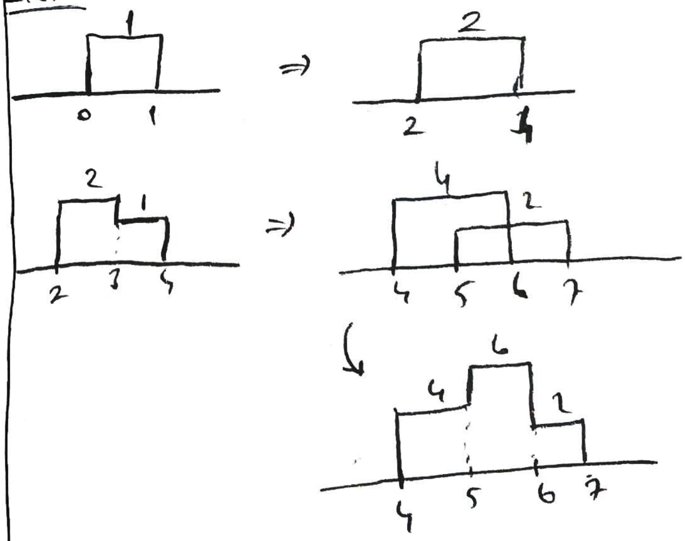
Linearity: $a x_i(t) \longrightarrow a y_i(t)$
$$x_1(t) \longrightarrow y_1(t)$$
$$x_2(t) \longrightarrow y_2(t)$$
$$x_1(t)+x_2(t) \longrightarrow y_1(t)+y_2(t)$$

Time Invariance
$$x(t) \longrightarrow y(t)$$
$$x(t-t_0) \rightarrow y(t-t_0)$$

Ex:



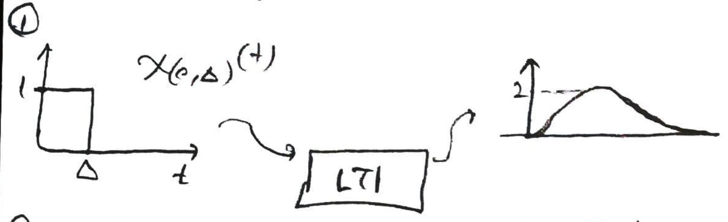
Ex:



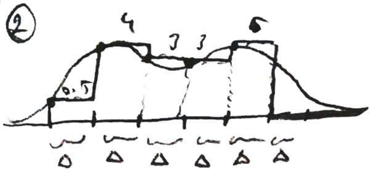
Question about "generality":

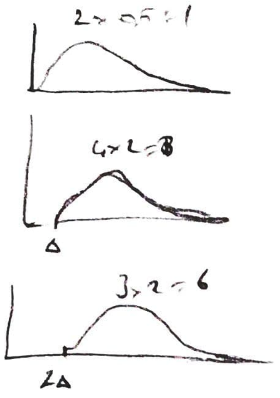
Assume we have LTI system. Can I compute its approximate output to arbitrary inputs after just a single experiment with the system?

①



②





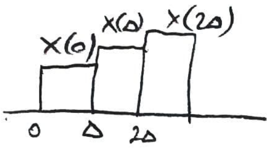
yer işte bu farklı yer parçalardı. Parçaları, oluştur[illegible] yerleri,

+

$$x_\Delta(t) = \sum_{n=-\infty}^{\infty} x(n\Delta)\, \chi_{(0,\Delta)}(t-n\Delta)$$

$$\Downarrow$$

$$= \dots x(0)\, \chi_{(0,\Delta)}(t) + x(\Delta)\, \chi_{(0,\Delta)}(t-\Delta) + \dots$$



LTI Systems for Discrete Case

Linearity:

$$x_1[n] \rightarrow y_1[n]$$
$$x_2[n] \rightarrow y_2[n]$$

$$a_1 x_1[n] + a_2 x_2[n] \rightarrow a_1 y_1[n] + a_2 y_2[n]$$

Time Invariance:

$$x[n] \rightarrow y[n]$$
$$x[n-n_0] \rightarrow y[n-n_0] \quad n_0 \in \mathbb{Z}$$

Q1: Assume LTI system

$\underline{1}\ 2 \longrightarrow \underline{4}\ 5$

$\underline{1}\ 3\ 2 \longrightarrow$

$1\ 2 \rightarrow \underline{4}\ 5$

$+\underline{0}\ 1\ 2 \rightarrow \underline{0}\ 4\ 5$

$\underline{1}\ 3\ 2 \rightarrow \underline{4}\ 9\ 5$

Q: LTI [illegible]

$\underline{1}\ 4\ 4 \longrightarrow ?$

$1\ 2 \longrightarrow \underline{4}\ 5$

$\times 2\ [0\ 1\ 2] \rightarrow [0\ 4\ 5] \times 2$

$\underline{1}\ 4\ 4 \rightarrow \underline{4}\ 13\ 10$

Q:

$\underline{1}\ 0\ -4 \rightarrow ?$

$1\ 2 \longrightarrow 4\ 5$

$-2\times [0\ 1\ 2]\quad (0\ 4\ 5]\ -2\times$

$\underline{1}\ 0\ -4 \rightarrow 4\ -3\ -10$

Q: $7\ 8 \rightarrow ?$

Best test function: $\underline{1}$

$\underline{1}$: ...000000001 00000...

$\underline{1} \longrightarrow \underline{4}\ -1\ 2$

$3\underline{2}\ 1\ 5 \longrightarrow ?$

$3\times (4\ \underline{-1}\ 2)$
$2\times (\underline{4}\ -1\ 2)$
$1\times (\underline{0}\ 4\ -1\ 2)$
$5\times (\underline{0}\ 0\ 4\ -1\ 2)$

$12\ \underline{-3}\ 6$
$8\ -2\ 4$
$0\ 4\ -1\ 2$
$0\ 0\ 20\ -10\ 10$

$12\ \underline{5}\ 8\ 23\ -18\ 10$

In discrete systems, if I know the response to $\underline{1}$, I know the response to any input $x[n]$

$\times$

$\underline{1} \rightarrow \dots h[-1]\ h[0]\ h[1] \dots$

$x[n] \rightarrow ?$

$x[-1] \rightarrow x[-1]\, h[k+1]$
$x[0] \rightarrow x[0]\, h[k]$
$+\ x[1] \rightarrow x[1]\, h[k-1]$

$$x[n] \rightarrow \sum_{k=-\infty}^{\infty} x[n]\, h[k-n]$$

systems (LTI)

continous ↙ ↘ Discrete

Date: 04.05.2021

Discrete (LTI)

$x[n] \longrightarrow \boxed{\text{system}} \longrightarrow y[n]$

Q:

$\underline{1} \longrightarrow \text{system} \longrightarrow h[n]$

$2\ \underline{3}\ 1\ 5 \longrightarrow \text{system} \longrightarrow 2h[n+1] + 3h[n] + h[n-1] + 5h[n-2]$

Q: $\underline{1} \longrightarrow \text{system} \longrightarrow \underline{0}\ 0\ 1\ 2\ 1$

$2\ \underline{3}\ 1\ 5 \longrightarrow \text{system} \longrightarrow$

$$\begin{array}{cccccccc}
 & 0 & \underline{0} & 2 & 4 & 2 & & \\
 & & \underline{0} & 0 & 3 & 6 & 3 & \\
 & & 0 & 0 & 0 & 1 & 2 & 1 \\
+ & & \underline{0} & 0 & 0 & 5 & 10 & 5 \\
\hline
 & & \underline{0} & 2 & 7 & 9 & 10 & 11\ 5
\end{array}$$

kronecker delta = impulse

General Question:

$\underline{1} \longrightarrow \boxed{LTI} \longrightarrow h[n]$ → impulse response

→ If I know the impulse response of an LTI system, I know everything

$x[n] \rightarrow \boxed{LTI} \rightarrow\ ?$

$$\cdots + x[-1]h[n+1] + x[0]h[n] + x[1]h[n-1] + \cdots$$

$$= \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] \Rightarrow \text{convolution sum}$$

Convolution Sum

$\underline{1} \longrightarrow LTI \longrightarrow h[n]$

$$x[n] \rightarrow LTI \rightarrow \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{k=-\infty}^{\infty} x[n-k]\,h[k]$$

example:

$\underline{1} \rightarrow \square \rightarrow \underline{0}\ 0\ 1\ 2\ 1$ $h[n]$

$x[k]$

$2\ \underline{3}\ 1\ 5 \rightarrow \square \rightarrow 0\ 2\ 7\ 9\ 10\ 5$

$$\begin{array}{cccccccccc}
 & & & 2 & \underline{3} & 1 & 5 & & & \\
1 & 2 & 1 & 0 & \underline{0} & & & \rightarrow & \underline{0} \\
 & 1 & 2 & 1 & 0 & 0 & & \rightarrow & 2 \\
 & & 1 & 2 & 1 & & & \rightarrow & 7 \\
 & & & 1 & 2 & 1 & & \rightarrow & 9 \\
 & & & & 1 & 2 & 1 & \rightarrow & 10 \\
 & & & & & 1 & 2\ 1 & \rightarrow & 5
\end{array}$$

→ bununla satırları çarpıtıyoruz

$$\text{"}\ \sum x[k]\,h[n-k]$$

example:

$\underline{1} \longrightarrow \square \longrightarrow \underline{0}\ 1\ 1\ -1\ 2$

$2\ 2\ \underline{0}\ 1\ -1 \longrightarrow \square \longrightarrow\ ?$

$y[0]$

$$\left.\begin{array}{ccccccc}
 & & \underline{0} & 1 & 1 & -1 & 2 \\
-1 & 1 & \underline{0} & 2 & 2 & &
\end{array}\right] \longrightarrow 4$$

$[-1]$

$$\left.\begin{array}{cccccccc}
 & & & \underline{0} & 1 & 1 & -1 & 2 \\
-1 & 1 & 0 & \underline{2} & 2 & & &
\end{array}\right] \longrightarrow 2$$

$[-2]$

$$\left.\begin{array}{ccccccccc}
 & & & & 0 & 1 & 1 & -1 & 2 \\
-1 & 1 & 0 & 2 & 2 & & & &
\end{array}\right] \longrightarrow 0$$

$n=1$

$$\left.\begin{array}{cccccc}
 & 0 & 1 & 1 & -1 & 2 \\
-1 & 1 & 0 & 2 & 2 &
\end{array}\right] \rightarrow 0$$

$n=2$

$$\left.\begin{array}{ccccc}
0 & 1 & 1 & -1 & 2 \\
-1 & 1 & 0 & 2 & 2
\end{array}\right] \rightarrow 3$$

$n=3$

$$\left.\begin{array}{cccccc}
0 & 1 & 1 & -1 & 2 & \\
 & -1 & 1 & 0 & 2 & 2
\end{array}\right] 4$$

$n=4$

$$\left.\begin{array}{ccccccc}
0 & 1 & 1 & -1 & 2 & & \\
 & & -1 & 1 & 0 & 2 & 2
\end{array}\right] -2$$

$n=5$

$$\left.\begin{array}{cccccccc}
0 & 1 & 1 & -1 & 2 & & & \\
 & & & -1 & 1 & 0 & 2 & 2
\end{array}\right] 3$$

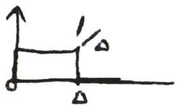
$n=6$

$$\left.\begin{array}{ccccccccc}
0 & 1 & 1 & -1 & 2 & & & & \\
 & & & & -1 & 1 & 0 & 2 & 2
\end{array}\right] -2$$

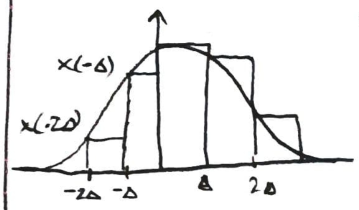
$$\boxed{2\quad \underline{4}\quad 0\quad 3\quad 4\quad -2\quad 3\quad -2}$$

Continous LTI Systems

$$\frac{x_{(0,\Delta)}(t)}{\Delta} \longrightarrow LTI \longrightarrow h_\Delta(t)$$



$x(t) \rightarrow LTI \rightarrow\ ?$



$$x(t) = x(-2\Delta)\cdot\frac{x_{(0,\Delta)}(t+2\Delta)}{\Delta}\cdot\Delta + x(-\Delta)\cdot\frac{x_{(0,\Delta)}(t+\Delta)}{\Delta}\cdot\Delta + x(0)\cdot\frac{x_{(0,\Delta)}(t)}{\Delta}\cdot\Delta + \ \vdots$$

$$y(t) \cong x(-2\Delta)h(t+2\Delta)\Delta + x(-\Delta)h(t+\Delta)\Delta + x(0)h(t)\Delta + x(\Delta)h(t-\Delta)\Delta + x(2\Delta)h(t-2\Delta)\Delta$$

$$x(t) \cong \sum_{k=-\infty}^{\infty} x(k\Delta)\,\frac{x_{(0,\Delta)}(t-k\Delta)}{\Delta}\,\Delta$$

$$y(t) \cong \sum_{k=-\infty}^{\infty} x(k\Delta)\,h(t-k\Delta)\,\Delta$$

$$\underset{\Delta\to 0}{x(t)} = \int_{s=-\infty}^{\infty} x(s)\,\delta(t-s)\,ds = x(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(s)\,h(t-s)\,ds$$

$x[n] \longrightarrow LTI \rightarrow \sum h[k]\,x[n-k]$

$x(t) \rightarrow LTI \rightarrow \int h(s)\,x(t-s)\,ds$

Convolution

Date: 06.05.2021

f we have LTI system ⇒ We have an impulse response $h[n], h(t)$

Discrete

$x[n] \rightarrow \boxed{\quad} \rightarrow y[n]$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$$

$1 \rightarrow \boxed{\quad} \rightarrow h[n]$

$\delta[n]$

continuous

$x(t) \rightarrow \boxed{\quad} \rightarrow y(t)$

$$y(t) = \int_{-\infty}^{\infty} h(s)\,x(t-s)\,ds$$

$\delta(t) \rightarrow \boxed{\quad} \rightarrow h(t)$

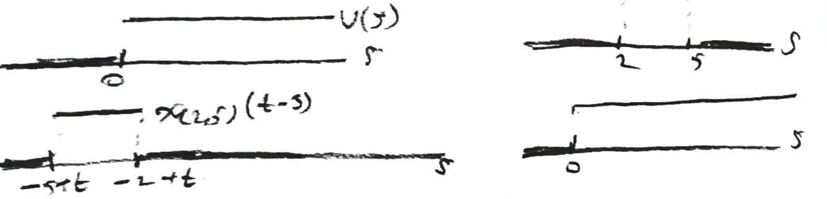
convolution operation is denoted by $*$

$(1\ \underline{2}\ 3) * (2\ \underline{1}\ 5)$

Example of continuous convolution:

1) Convolve $\chi_{(2,5)}t$ with $u(t)$

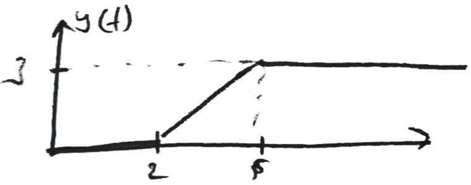
$y(t) = \chi_{(2,5)}(t) * u(t)$



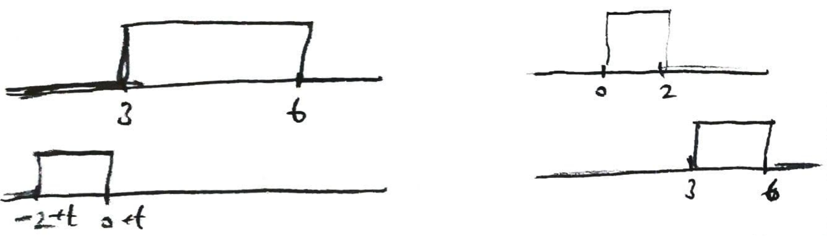
$\chi_{(2,5)}(s)$
$u(s)$
s
0
2
5
s
$\chi_{(2,5)}(t-s)$
$-5+t$
$-2+t$
s
0
s

$$\int_{-\infty}^{\infty} u(s)\,\chi_{(2,5)}(t-s)\,ds$$

$y(t)=0$	$t<2$
$y(t)=-2+t$	$2<t<5$
$y(t)=3$	$5<t$

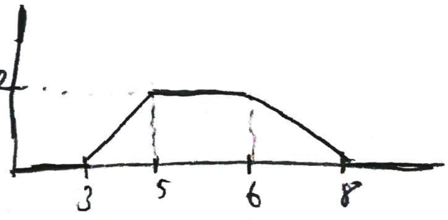


2) $y(t) = \chi_{(0,2)}(t) * \chi_{(3,6)}(t)$

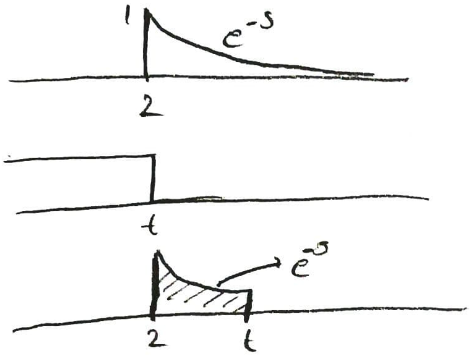


3
6
$-2+t$
$0+t$
0
2
3
6

$y(t)=0$	$t<3$ (dışarıda)
$y(t)=t-3$	$3<t<5$ (ortada)
$y(t)=2$	$5<t<6$ (içeride)
$y(t)=8-t$	$6<t<8$ (ortada)
$y(t)=0$	$8<t$ (dışarıda)



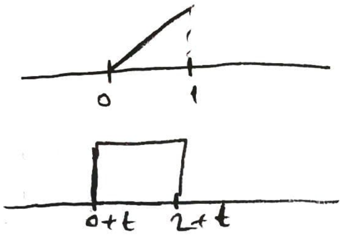
3) $y(t) = e^{-t}u(t-2) * u(t)$



$$\int_2^t e^{-s}\,ds = \frac{e^{-s}}{-1}\Big|_2^t = e^{-2} - e^{-t}$$

$$y(t) = \begin{cases} 0, & t<2 \\ e^{-2}-e^{-t}, & 2<t \end{cases}$$

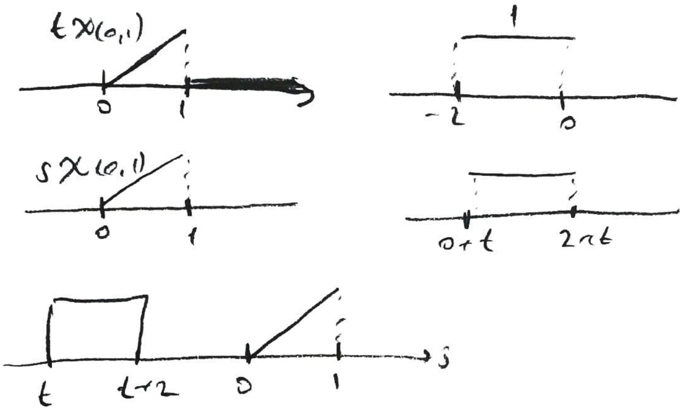
4) $y(t) = t\,\chi_{(0,1)}(t) * \chi_{(-2,0)}(t)$



0	$t<-2$
$\int_0^{t+2} s\,ds$	$-2<t-1$
$\int_0^1 s\,ds$	$-1<t<0$
0	$1<t$

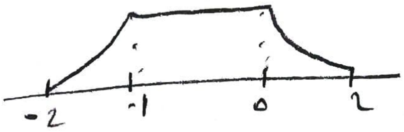
Date: 11.05.2021

5) $t\,\chi_{(0,1)}(t) * \chi_{(-2,3)}(t)$

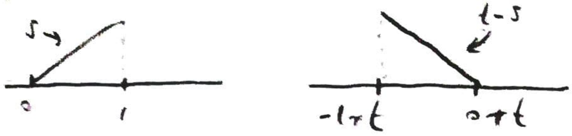


$t\chi_{(0,1)}$
0
1
1
-2
0
$s\chi_{(0,1)}$
0
1
$0+t$
$2+t$
t
$t+2$
0
1
s

	0	$t<-2$
$\frac{(t+2)^2}{2}$	$\int_0^{t+2} s\,ds$	$-2<t<-1$
$\left(\frac{1}{2}\right)^2$	$\int_0^1 s\,ds$	$-1<t<0$
$\frac{1}{2}-\frac{t^2}{2}$	$\int_t^1 s\,ds$	$0<t<1$
	0	$1<t$



8) $t\chi_{(0,1)}(t) * t\chi_{(0,1)}(t)$



$$\begin{cases} 0 & t<0 \\ \int_0^t s(t-s)ds & 0<t<1 \\ \int_{t-1}^{t} s(t-s)ds & 1<t<2 \\ 0 & 2<t \end{cases}$$

9) $e^{t}u(t) * t^2\chi_{(0,1)}(t)$

not solved

Discrete LTI SYSTEMS

= Linear Constant coeff. Difference Eq.

$$5y[n]+2y[n-1]-y[n-2]=x[n]$$

$$x[n] \longrightarrow \boxed{\text{system}} \longrightarrow y[n]$$

$\delta[n]$ + initial conditions

initial condition (always 0 for our case)

$y[-2]=0$ $\quad y[0]=1$

$y[-1]=0$ $\quad y[2]=$

$x[n]=\delta[n]$

$$\left.\begin{aligned} 5y[0]+2y[-1]-y[-2] &= x[0] \\ 5y[1]+2y[0]+y[-1] &= x[1] \\ 5y[2]+2y[1]+y[0] &= x[2] \\ &\vdots \end{aligned}\right\} \begin{aligned} &x[n] \\ &\delta[n] \end{aligned}$$

$y[0]=\frac{1}{5} \quad y[1]=-\frac{2}{25} \quad y[2]=\frac{1}{5}\left[2\frac{2}{25}-5\frac{1}{5}\right]$

"The impulse response is infinite"

$$x[n] \longrightarrow \boxed{ay[n]+by[n-1]=x[n]} \longrightarrow y[n]$$

First Order LTI Systems = first order constant coefficient difference equations

---

$ny[n]+5y[n-1]=x[n] \rightarrow$ X 'dış!!!!

$ay[n]+by[n-1]=x[n] \qquad x[n]\rightarrow\boxed{}\rightarrow y[n]$

Find Impulse Response :

ex: $ay[n]+by[n-1]=\delta[n] \qquad y[-1]=0$

$n=0 \quad ay[0]+by[-1]=1 \qquad y[0]=\frac{1}{a}$

$n=1 \quad ay[1]+b\frac{1}{a}=0 \qquad y[1]=-\frac{b}{a^2}$

$n=2 \quad ay[2]+b\left(\frac{-b}{a^2}\right)=0 \qquad y[2]=\frac{b^2}{a^2}$

$$y[n]=(-1)^n\frac{b^n}{a^{n+1}}$$

ex:

if $x[n]=u[n]$

$$y[n]=u[n] * (-1)^n\frac{1}{a}\left(\frac{b}{a}\right)^n$$

$$\frac{1}{a}-\frac{b}{a^2}+\frac{b^2}{a^3}-\frac{b^3}{a^4}\cdots$$

$$\begin{aligned} S &= 1+\alpha+\alpha^2+\alpha^3+\cdots \qquad \alpha<1 \\ -\ \alpha S &= \alpha+\alpha^2+\alpha^3+\cdots \\ \hline S-\alpha S &= 1 \qquad S=\frac{1}{1-\alpha} \end{aligned}$$

$$\begin{aligned} S &= 1+\alpha+\alpha^2+\cdots\ \alpha^n \qquad \alpha \text{ can be anything} \\ \alpha S &= \alpha+\alpha^2+\alpha^3\cdots \\ \hline S-\alpha S &= 1-\alpha^{n+1} \qquad S=\frac{1-\alpha^{n+1}}{1-\alpha} \end{aligned}$$

$$y[n]=\sum_{m=0}^{n}\frac{1}{a}\left(-\frac{b}{a}\right)^m=\frac{1}{a}\sum_{m=0}^{n}\left(-\frac{b}{a}\right)^m$$

$$=\frac{1}{a}\,\frac{1-\left(\frac{-b}{a}\right)^{n+1}}{1-\left(-\frac{b}{a}\right)}u[n]$$

$$x[n]\rightarrow\boxed{}\rightarrow y[n]$$

$$u[n]\rightarrow\boxed{}\rightarrow\frac{1}{a}\,\frac{1-\left(\frac{-b}{a}\right)^{n+1}}{1-\left(-\frac{b}{a}\right)}u[n]$$

$$u[n]+3u[n-4]\rightarrow\boxed{}\longrightarrow\frac{1}{a}\,\frac{1-\left(\frac{-b}{a}\right)^{n-3}}{1-\left(-\frac{b}{a}\right)}u[n-4]$$

$$+\frac{1}{a}\,\frac{1-\left(-\frac{b}{a}\right)^{n+1}}{1-\left(-\frac{b}{a}\right)}u[n]$$

ex: (Same system)

$$\chi_{(0,5)}(t)\longrightarrow\boxed{}\longrightarrow ?$$

$$u[n]-u[n-6]\rightarrow\boxed{}\rightarrow\frac{1}{a}\,\frac{1-\left(\frac{-b}{a}\right)^{n+1}}{1+\frac{b}{a}}u[n]$$

$$\frac{1}{a}\,\frac{1-\left(\frac{-b}{a}\right)^{n-5}}{1+\frac{b}{a}}u[n-6]$$

Example of Continuous LTI System . LCC diff eq.

$$\frac{dy(t)}{dt} + ay = x(t) \qquad x(t) \to \boxed{} \to y(t)$$

$$\frac{dh}{dt} + ay = \delta(t) \Rightarrow \text{find } h$$

for general input $x(t)$

$$y(t) = \int_{-\infty}^{\infty} h(t-s)\, x(s)\, ds$$

Date: 18.05.2021

Discrete LTI Systems

Discrete LTI system = LCC difference eq.

$$y[n] + a_1 y[n-1] + a_2 y[n-2] \dots = x[n] + b_1 x[n-1] + b_2 x[n-2] \dots$$

$$x[n] \to \boxed{\text{system}} \to y[n]$$

everything starts at time = 0

$y[n] = 0 \quad n < 0 \qquad x[n] = 0 \quad n < 0$

we have done $y[n] + a y[n-1] = x[n]$

summary ↑

Continuous LTI Systems

= LCC differential eq.

$$\frac{d^n y}{dt^n} + a_1 \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_n y = x + b_1 \frac{dx}{dt} + b_2 \frac{d^2 x}{dt^2} + \dots b_n \frac{d^n x}{dt^n}$$

$$\left(\frac{dy}{dt}\right)^2 + 3e^y = x + \cos\left(\frac{dx}{dt}\right) \Rightarrow \text{not linear}$$

$t \frac{dy}{dt} + y = x$ not constant coeff.

The case will be interested

$$\frac{dy(t)}{dt} + a y(t) = x(t)$$

① $\frac{dy}{dt} + ay = 0 \qquad e^{at}\left(\frac{dy}{dt} + ay\right) = e^{at} \cdot 0 = 0$

$$\Rightarrow e^{at} \frac{dy}{dt} + a e^{at} y = 0$$

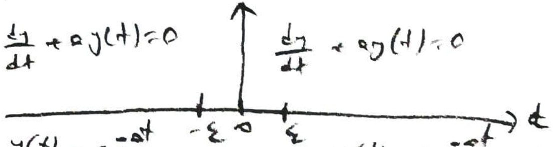
$$\Rightarrow \frac{d}{dt}\left(e^{at} y\right) = 0$$

$$\Rightarrow e^{at} y = C \quad \swarrow \text{constant}$$

$$\Rightarrow y(t) = C \cdot e^{-at}$$

we have to find the impulse response of a LTI system to extract all the info about it

$$\frac{dy(t)}{dt} + y(t) = \delta(t)$$



$\frac{dy}{dt} + a y(t) = 0 \qquad \frac{dy}{dt} + a y(t) = 0$

$t < -\varepsilon \quad y(t) = c_0 e^{-at} \qquad y(t) = c_1 e^{-at} \quad t > \varepsilon$

c may be different

I assume that before the "peak" of dirac delta the system is at rest $\Rightarrow y(t) = 0$

$\Rightarrow c_0 = 0$

let us consider the equation in $(-\varepsilon, \varepsilon)$

$$\frac{dy}{dt} + ay = \delta(t)$$

The two side must balance, and δ must be finite.

→ $y(t)$ must behave like unit step in $t \in [-\varepsilon, \varepsilon]$, i.e.,

$y(-\varepsilon) = y(0^-) = 0$

$y(+\varepsilon) = y(0^+) = 1$

$y(t) = 0 \quad t < -\varepsilon$

$y(t) = c_1 e^{-at} \quad t > \varepsilon$

$y(-\varepsilon) = 0$

$y(+\varepsilon) = 1$

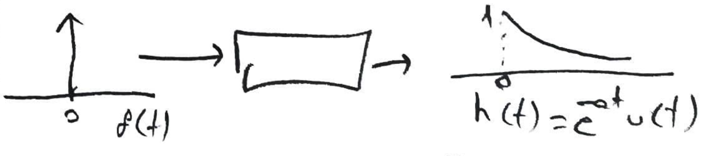
$y(+\varepsilon) = c_1 e^{-a\varepsilon} = 1$

$\Rightarrow c_1 = 1$

$$\left.\begin{array}{ll} y(t) = 0 & t < 0 \\ y(t) = e^{-t} & t > 0 \end{array}\right\} y(t) = e^{-at} u(t)$$

$h(t) = e^{-at} u(t)$

Result: impulse response of LTI system $\frac{dy}{dt} + ay = x$ is $h = e^{-at} u(t)$



$$x(t) \to \boxed{} \to y(t) = \int_{-\infty}^{\infty} e^{-as} u(s)\, x(t-s)\, ds$$

$$= \int_{-\infty}^{\infty} e^{-a(t-s)} u(t-s)\, x(s)\, ds$$

$$x(t) * y(t) = \int_{-\infty}^{\infty} x(t-s)\, y(s)\, ds = \int_{-\infty}^{\infty} x(s)\, y(t-s)\, ds$$

fourier Transforms Date: 2005/2020

continous	Discrete
CFT	DFT

$x(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} X(\omega)$ | $x[n] \underset{IFT}{\overset{FT}{\rightleftharpoons}} X(\Omega)$

time domain | frequency domain | time domain | frequency domain

Continous FT's:

$i = j = \sqrt{-1}$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

w → frequency

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

degree artarsa escillete hizi artar

$\omega = 2\pi f$

$$x(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t)\cos\omega t\, dt - j\int_{-\infty}^{\infty} x(t)\sin\omega t\, dt$$

$\rightarrow 0$ $\rightarrow 0$

w goes to infinity.

$$2\pi x(t) = \sum_{k=-\infty}^{\infty} X(k\Delta\omega) e^{jk\Delta\omega} \cdot \Delta\omega = \sum_{k=-\infty}^{\infty} a_k e^{jk\Delta\omega}$$

"A fourier transform writes a signal x(t) as a "sum/integral" of complex exponentials

$$x(t) = \underline{\underline{2}} e^{j2\pi \underline{5} t} + \underline{\underline{100}} e^{j2\pi \underline{40} t} + \underline{\underline{2}} e^{j2\pi \underline{500} t}$$

Ex:

Find the FT of $e^{-\alpha t} u(t) = x(t)$

$$x(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \boxed{e^{-\alpha t} u(t)}\, e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-\alpha t - j\omega t} u(t)\, dt$$

$\boxed{\alpha > 0}$

$$= \int_{0}^{\infty} e^{-\alpha t - j\omega t} dt = \left.\frac{e^{-(\alpha + j\omega)t}}{-(\alpha + j\omega)}\right|_{0}^{\infty}$$

$$= \cancel{\frac{e^{-(\alpha + j\omega)\infty}}{-(\alpha + j\omega)}} \rightarrow 0 - \frac{e^{-(\alpha + j\omega)0}}{-(\alpha + j\omega)} = \frac{1}{\alpha + j\omega}$$

$$\downarrow$$

$$\frac{e^{-\alpha\infty} \cdot e^{-j\omega\infty}}{-(\alpha + j\omega)}$$

$$\boxed{FT[e^{-\alpha t} u(t)] = \frac{1}{\alpha + j\omega}}$$

$\alpha > 0$

Ex: $FT(e^{-a|t|})$ $a > 0$

$$= \int_{-\infty}^{\infty} e^{-a|t|} e^{-j\omega t} dt = \int_{-\infty}^{0} e^{at} e^{-j\omega t} dt + \int_{0}^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \left.\frac{e^{(\alpha - j\omega)t}}{\alpha - j\omega}\right|_{\infty}^{0} + \frac{1}{\alpha + j\omega} = \frac{1}{\alpha - j\omega} + \frac{1}{\alpha - j\omega}$$

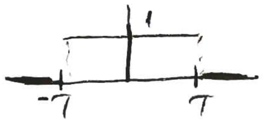
$$= \frac{2\alpha}{\alpha^2 + \omega^2}$$

contribution of 5 Hz'i

$$\frac{2\alpha}{\alpha^2 + (2\pi 5)^2}$$

// frequency amplitude // contribution verir.

Ex: $x_{(-T,T)}(t)$



$$x(\omega) = \int_{-\infty}^{\infty} x_{(-T,T)}(t) e^{-j\omega t} dt$$

$$= \int_{-T}^{T} e^{-j\omega t} dt = \left.\frac{e^{-j\omega t}}{-j\omega}\right|_{-T}^{T}$$

$$= \frac{e^{-j\omega t}}{-j\omega} - \frac{e^{-j\omega(-T)}}{-j\omega} = \frac{e^{j\omega t} - e^{-j\omega t}}{j\omega}$$

$$\left.\begin{array}{l} e^{j\theta} = \cos\theta + i\sin\theta \\ e^{-j\theta} = \cos(-\theta) + i\sin(-\theta) \end{array}\right\} \rightarrow \begin{array}{l} e^{j\theta} = \cos\theta + i\sin\theta \\ -e^{-j\theta} = -\cos\theta + i\sin\theta \end{array}$$

$$\boxed{\frac{e^{j\theta} - e^{-j\theta}}{2} = \sin\theta}$$

Bunu olarak:

$$\frac{2 \cdot e^{j\omega t} - e^{-j\omega t}}{2 \cdot j\omega} \Rightarrow \frac{2\sin(\omega T)}{\omega}$$

Ex: $FT[\delta(t)] = ?$

$$\int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

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Ex: $FT[\delta(t-c)] = ?$

$$\int_{-\infty}^{\infty} \delta(t-c) e^{-j\omega t} dt = e^{-j\omega c}$$

Ex: $FT[e^{jct}]$

$$\int_{-\infty}^{\infty} e^{jct} \cdot e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{j(c-\omega)t} dt$$

$$= \left.\frac{e^{j(c-\omega)t}}{c-\omega}\right|_{\infty}^{\infty} = \frac{e^{j(c-\omega)\infty} - e^{j(c-\omega)(-\infty)}}{c-\omega}$$

next page →

Other way: $\lim_{T\to\infty}\int_{-T}^{T} e^{j(c-\omega)t}\,dt = \frac{e^{j(c-\omega)t}}{j(c-\omega)}\Big|_{-T}^{T}$

$\uparrow \lim_{T\to\infty}$

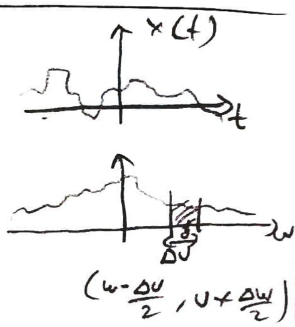
$$= \lim_{T\to\infty} \frac{e^{j(c-\omega)T} - e^{-j(c-\omega)T}}{j(c-\omega)}$$

$$= \lim_{T\to\infty} \frac{2\sin(c-\omega)T}{c-\omega} = \delta(c-\omega)$$

Date: 25/05/2011

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\,e^{j\omega t}\,dt$$



$$\boxed{\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}}$$

bun
untine
songle de
klew sin

Ex: FT(1)

$$\int_{-\infty}^{\infty} 1\,e^{-j\omega t}\,dt = \cancel{\frac{e^{-j\omega t}}{j\omega}\Big|_{-\infty}^{\infty} = \frac{e^{-j\omega\infty}}{j\omega} - \frac{e^{-j\omega(-\infty)}}{j\omega}}$$

$$FT(1) = \lim_{T\to\infty}\int_{-T}^{T} e^{-j\omega t}\,dt = \lim_{T\to\infty} \frac{2\sin(\omega T)}{\omega} = 2\pi\delta(\omega)$$

Ex: FT$(\bar{e}^{j\alpha t})$

$$\lim_{T\to\infty}\int_{-T}^{T} e^{j\alpha t}e^{-j\omega t} = \lim_{T\to\infty} \frac{e^{-j(\alpha+\omega)t}}{-j(\alpha+\omega)}\Big|_{-T}^{T}$$

$$= \lim_{T\to\infty} \frac{e^{j(\alpha+\omega)T} - e^{-j(\alpha+\omega)T}}{2j(\alpha+\omega)}$$

$$= \lim_{T\to\infty} \frac{2[\sin(\alpha+\omega)T]}{(\alpha+\omega)} = 2\pi\delta(\alpha+\omega)$$

Ex: $\sum_{n=-\infty}^{\infty} \delta(t-nT)$

$$\int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t-nT)\,e^{-j\omega t}\,dt = \sum_{n=-\infty}^{\infty}\int_{-\infty}^{\infty} \delta(t-nT)\,e^{-j\omega t}\,dt$$

$$= \sum_{n=-\infty}^{\infty} e^{-j\omega nT}$$

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Ex: FT$(N(t))$

$$\int_{-\infty}^{\infty} N(t)\,e^{-j\omega t}\,dt = \int_{0}^{\infty} e^{-j\omega t}\,dt$$

$$\lim_{T\to\infty}\int_{0}^{T} e^{-j\omega t}\,dt = \lim_{T\to\infty} \frac{e^{-j\omega t}}{-j\omega}\Big|_{0}^{T}$$

$$= \lim_{T\to\infty}\left(\frac{1}{j\omega} - \frac{e^{-j\omega t}}{j\omega}\right) = \frac{1}{j\omega} - \lim_{T\to\infty}\frac{e^{-j\omega t}}{j\omega}$$

$$= \frac{1}{j\omega} - \lim_{T\to\infty} \frac{\cos(\omega T) - j\sin(\omega T)}{j\omega}$$

$$= \frac{1}{j\omega} + \cancel{j\lim_{T\to\infty}\frac{\cos(\omega T)}{\omega}} + \underbrace{\lim_{T\to\infty}\frac{\sin(\omega T)}{\omega}}_{\pi\delta(\omega)}$$

$$= \frac{1}{j\omega} + \pi\delta(\omega)$$

Properties of CFT

1) Linear: $FT(aX(t)+by(t)) = aFT(x(t)) + bFT(y(t))$

$$\int_{-\infty}^{\infty} (ax(t)+by(t))e^{-j\omega t}\,dt = a\underbrace{\int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt}_{X(\omega)} + b\underbrace{\int_{-\infty}^{\infty} y(t)e^{-j\omega t}\,dt}_{Y(\omega)}$$

2) Symmetry: for $x(t)$ real

$$X(-\omega) = X^*(\omega)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\,e^{-j\omega t}\,dt$$

$$X^*(\omega) = \int_{-\infty}^{\infty} x(t)\,e^{j\omega t}\,dt$$

$$= X(-\omega)$$

$(e^{j\omega t})^* = [\cos(\omega t) + j\sin(\omega t)]^*$

$= e^{-j\omega t}$

3) Time shifting: $x(t) \rightarrow X(\omega)$

$x(t-t_0) \rightarrow e^{j\omega(t_0)}X(\omega)$

$FT[u(t)] \rightarrow \frac{1}{j\omega} + \pi\delta(\omega)$

$FT[u(t-5)] \rightarrow \frac{e^{-j\omega 5}}{j\omega} + \pi e^{-j\omega 5}\delta(\omega) = \frac{e^{-j\omega 5}}{j\omega} + \pi\delta(\omega)$

Proof: $FT[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = X(\omega)$

$FT[x(t-t_0)] = \int_{-\infty}^{\infty} x(t-t_0) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(u) e^{-j\omega(u+t_0)} du$

$u = t - t_0 \quad dt = du$

$= \int_{-\infty}^{\infty} x(u) e^{-j\omega u} \cdot e^{-j\omega t_0} du = e^{-j\omega t_0} \int_{-\infty}^{\infty} x(u) e^{-j\omega u} du$

$= e^{-j\omega t_0} X(\omega)$

"bu önemli"
time shifting proof

Date: 27.05.2021

4) Differentiation & Integration Prop.

$x(t) \to X(\omega) \quad : \quad \frac{d\,x(t)}{dt} \to ? \quad \int_{-\infty}^{t} x(s) ds \to ?$

$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad \frac{d}{dt} x(t) = \frac{d}{dt} \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$

$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot \frac{d}{dt} (e^{j\omega t}) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)\, i\omega\, e^{i\omega t} d\omega$

$= \frac{1}{2\pi} \int_{-\infty}^{\infty} i\omega\, X(\omega)\, e^{i\omega t} d\omega$

$\underbrace{\qquad\qquad}_{}$ derivative

$$\boxed{\frac{d\,x(t)}{dt} \to j\omega X(\omega)}$$

Differentiation in time domain corresponds to multiplication with $j\omega$ in freq. domain.

Integration :

$x(t) \Longleftrightarrow X(\omega) \quad \int_{-\infty}^{t} x(s) ds \Longleftrightarrow ?$

"ıvır zıvır olan kelimelerin ordinden"

General Rule:

$X(t) \longrightarrow X(\omega)$

$$\boxed{\int_{-\infty}^{t} x(s) ds \longrightarrow \pi \delta(\omega) X(0) + \frac{1}{j\omega} X(\omega)}$$

Quiz Sorusu: $f(t) * \delta(t-5)$

$= \int_{-\infty}^{\infty} f(t-s) \delta(s-5) ds = f(t-5)$

Not: $f(t) * \delta(t) = \int_{-\infty}^{\infty} f(s) \delta(t-s) ds = f(t)$

Ex: $\frac{d\,y(t)}{dt} + 5y(t) = x(t) \qquad x(t) \to \square \to y(t)$

$FT\left(\frac{d\,h(t)}{dt} + 5h(t) \right) = (\delta(t))\ FT$

$j\omega h(\omega) + 5h(\omega) = 1 \qquad e^{-t} u(t) = \frac{1}{j\omega + 1}$

$(j\omega + 1) h(\omega) = 1$

$h(\omega) = \frac{1}{j\omega + 1} \Rightarrow \underline{e^{-t} u(t) = h(t)}$

Ex: $(x+2)(x+3) = x^2 + 5x + 6$

$\frac{d^2 y}{dt^2} + 5\frac{dy}{dt} + 6y = \delta(t) \qquad$ FT of

$(j\omega)(j\omega) Y(\omega) + 5j\omega Y(\omega) + 6Y(\omega) = 1$

$(j\omega)^2 Y(\omega) + 5j\omega Y(\omega) + 6Y(\omega) = 1$

$Y(\omega) = \frac{1}{(j\omega)^2 + 5j\omega + 6} = \frac{1}{(j\omega+2)(j\omega+3)}$

$= \frac{A}{j\omega+2} + \frac{B}{j\omega+3}$ and so on, . . . $\quad A = 1$, $B = -1$

$= \frac{1}{j\omega+2} - \frac{1}{j\omega+3}$

$$\boxed{Y(t) = e^{-2t} u(t) - e^{-3t} u(t)}$$

kolayyy

5) Convolution Property:

$x(t) \to X(\omega)$
$y(t) \to Y(\omega)$

convolution in time is multiplication in freq.

$x(t) * y(t) \to X(\omega) \cdot Y(\omega)$

$x(t) \to \square \to y(t) = \int_{-\infty}^{\infty} x(s) h(t-s) ds = \int_{-\infty}^{\infty} x(t-s) h(s) ds$ } LTI in time

$X(\omega) \to \square \to Y(\omega) = H(\omega) X(\omega)$ } LTI in freq.

Ex:

$\cos(40t) \longrightarrow \boxed{h(t) = e^{-5t} u(t)} \to y(t) = \cos 40t * e^{-5t} u(t)$

$\longrightarrow \boxed{\frac{1}{j\omega + 5}}$

→ next page

$$\pi\delta(\omega-\omega_0)+\pi\delta(\omega+\omega_0) \rightarrow \boxed{\frac{1}{j\omega+5}} \rightarrow$$

$$\frac{\pi\delta(\omega-\omega_0)}{j\omega+5}+\frac{\pi\delta(\omega+\omega_0)}{j\omega+5}$$

in time: $\frac{\pi}{j\omega_0+5}\frac{e^{j\omega_0 t}}{2\pi}+\frac{\pi}{-j\omega_0+5}\frac{e^{-\omega_0 jt}}{2\pi}$

6) Scaling Property: $FT[f(t)] \rightarrow F(\omega)$

$FT[f(at)] \rightarrow \frac{1}{|a|}F\left(\frac{\omega}{a}\right)$

7) Modulation Property:

$$\begin{tikzcd} r(t) \arrow[d] & = & s(t) \arrow[d] & p(t) \arrow[d] \\ R(\omega) & & S(\omega) & P(\omega) \end{tikzcd}$$

$\rightarrow R(\omega)=\frac{1}{2\pi}S(\omega)*P(\omega)$

Fourier Transforms of Periodic Functions (Fourier Series)

$f(t+T)=f(t)$ T is the period

$$F(\omega)=\int_{-\infty}^{\infty}f(t)e^{-j\omega t}dt=\sum_{n=-\infty}^{\infty}\int_{nT}^{(n+1)T}f(t)e^{-j\omega t}dt$$

$$=\sum_{n=-\infty}^{\infty}\int_0^T f(t+nT)e^{-j\omega(t+nT)}dt$$

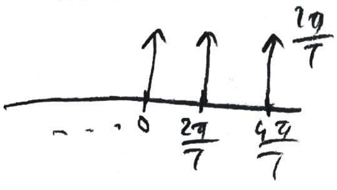
$$=\sum_{n=-\infty}^{\infty}\int_0^T f(t)e^{-j\omega(t+nT)}dt$$

"U gonna get"

$$=\sum_{n=-\infty}^{\infty}\left[\int_0^T f(t)e^{-j\omega t}dt\right]\cdot e^{-j\omega nT}$$

$$=\left[\int_0^T f(t)e^{-j\omega t}dt\right]\cdot\sum_{n=-\infty}^{\infty}e^{-j\omega nT}$$

Theorem: $\sum_{n=-\infty}^{\infty}e^{-j\omega nT}=\frac{2\pi}{T}\sum_{n=-\infty}^{\infty}\delta\left(\omega-n\frac{2\pi}{T}\right)$



$$=\lim_{N\to\infty}\sum_{n=-N}e^{-j\omega nT}=\lim_{N\to\infty}\left(e^{-j\omega NT}+e^{-j\omega(N-1)T}\ldots\ldots+e^{j\omega NT}\right)$$

$$=\lim_{N\to\infty}e^{-j\omega NT}\underbrace{\left(1+\ldots\ldots+e^{j\omega 2NT}\right)}_{\text{power series}}$$

$$=\lim_{N\to\infty}e^{-j\omega NT}\cdot\left(\frac{1-(e^{j\omega T})^{2N+1}}{1-e^{j\omega T}}\right)$$

Aside: $x^{-N}\cdot\frac{1-x^{2N+1}}{1-x}$ $x=e^{j\omega T}$

$$=\cancel{x^{-N}}\frac{\cancel{x^{\frac{2N+1}{2}}}\left(x^{-\frac{2N+1}{2}}-x^{\frac{2N+1}{2}}\right)}{\cancel{x^{1/2}}\left(x^{-1/2}-x^{1/2}\right)}$$

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$$=\frac{x^{-\frac{2N+1}{2}}-x^{\frac{2N+1}{2}}}{x^{-1/2}-x^{1/2}}$$

$$\Rightarrow\lim_{N\to\infty}\frac{(e^{j\omega T})^{-\frac{2N+1}{2}}-(e^{j\omega T})^{\frac{2N+1}{2}}}{(e^{j\omega T})^{-1/2}-(e^{j\omega T})^{1/2}}$$

$\Downarrow$

$$=\lim_{N\to\infty}\frac{\sin\left[\frac{2N+1}{2}\omega T\right]}{\sin\left[\frac{\omega T}{2}\right]}$$

"so u can learn"

sums check:

$$\sum_{n=-\infty}^{\infty}e^{-j\omega nT}=\lim_{N\to\infty}\frac{\sin\left[\frac{2N+1}{2}\omega T\right]}{\sin\left[\frac{\omega T}{2}\right]}$$

* when does $\sin\left[\frac{\omega T}{2}\right]\neq 0$:

$$\lim_{N\to\infty}\frac{\sin\left[\frac{2N+1}{2}\omega T\right]}{\sin\left(\frac{\omega T}{2}\right)}=0$$

* when does $\sin\left(\frac{\omega T}{2}\right)=0$:

$\frac{\omega T}{2}=k\pi$ $\omega=\frac{2k\pi}{T}$

$$\lim_{N\to\infty}\frac{\sin\left[\frac{2N+1}{2}\omega T\right]}{\sin\left[\frac{\omega T}{2}\right]}\cong\frac{\sin\left[\frac{2N+1}{2}\omega T\right]}{\frac{\omega T}{2}-k\pi}$$

$$\cong\frac{2}{T}\lim_{N\to\infty}\frac{\sin\left[\frac{2N+1}{2}\omega T\right]}{\omega-\frac{2k\pi}{T}}=2T\cdot\pi\delta\left(\omega-\frac{2k\pi}{T}\right)$$

Sum sums:

$$\sum_{n=-\infty}^{\infty}e^{-j\omega nT}=\lim_{N\to\infty}\frac{\sin\left[\frac{2N+1}{2}\omega T\right]}{\sin\left[\frac{\omega T}{2}\right]}=\frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\delta\left(\omega-\frac{2k\pi}{T}\right)$$

Result: $\sum_{n=-\infty} e^{-j\omega nT} = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2k\pi}{T}\right)$

$$\int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \left[\int_0^T f(t) e^{-j\omega T} dt\right]\left[\sum_{n=-\infty}^{\infty} e^{-j\omega n T}\right]$$

$$= \left[\int_0^T f(t) e^{-j\omega t} dt\right]\left[\frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\delta\left(\omega - \frac{2k\pi}{T}\right)\right]$$

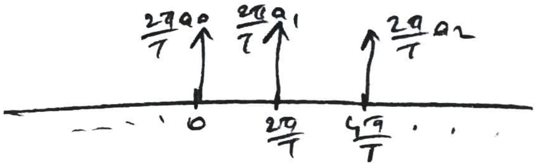
$$= \frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\left(\int_0^T f(t) e^{-j\omega t} dt\right)\delta\left(\omega - \frac{2k\pi}{T}\right)$$

$$= \frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\underbrace{\left[\int_0^T f(t) e^{-j\frac{2k\pi}{T}t} dt\right]}_{a_k}\delta\left(\omega - \frac{2k\pi}{T}\right)$$

$$= \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} a_k \delta\left(\omega - \frac{2k\pi}{T}\right)$$

→ compute $a_k = \int_0^T f(t) e^{-j\frac{2\pi}{T}t} dt$

→ compute $\sum_{-\infty}^{\infty} a_k \delta\left(\omega - \frac{2k\pi}{T}\right)$



Periodic FT

Date: 03.06.2021

if $x(t+T) = x(t)$

$X(\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} a_k \delta\left(\omega - k\frac{2\pi}{T}\right)$ $a_k = \int f(t) e^{-j\frac{2\pi k}{T}} dt$

Returning back to time:

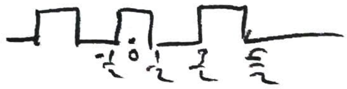
$$x(t) = \int_{-\infty}^{\infty} \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} a_k \delta\left(\omega - k\frac{2\pi}{T}\right) e^{j\omega t} d\omega$$

$$= \frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\int_{-\infty}^{\infty} a_k \delta\left(\omega - k\frac{2\pi}{T}\right) e^{j\omega t} d\omega$$

$$\boxed{= \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi k}{T}t}}$$ ⟹ Fourier series representation of $x(t)$

Ex: Take the FT of $\chi_{(-\frac{1}{2},\frac{1}{2})}(t-2n) = x(t)$

$a_k = \int_{-1}^{1} \chi_{(-1/2,1/2)} e^{-j\frac{2k\pi}{2}t} dt$



$a_k = \int_{-1/2}^{1/2} e^{-j\frac{2k\pi}{2}t} dt = \left.\frac{e^{-jk\pi t}}{-jk\pi}\right|_{-1/2}^{1/2}$

$a_k = \frac{e^{jk\pi/2} - e^{-jk\pi/2}}{jk\pi} = \boxed{\frac{2\sin\left(\frac{k\pi}{2}\right)}{k\pi} = a_k}$

continue: $X(\omega) = \frac{2\pi}{2}\sum_{k=-\infty}^{\infty}\frac{2\sin(k\pi/2)}{k\pi}\delta\left(\omega - \frac{k2\pi}{\pi}\right)$

FT ⟶

$x(t) = \frac{2\pi}{2}\sum_{k=-\infty}^{\infty}\frac{2\sin(k\pi/2)}{k\pi} e^{j\frac{2\pi k t}{2}}$

ilginç bilgi:

✓ $t^2\delta(t-5) = 25\delta(t-5)$ (yanlış olabilir

✓ $\int t^2\delta(t-5) = \int 5\delta(t-5) = 5$

Discrete FT's (DFT)

$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$ → continuous and periodic

$x[n] = \frac{1}{2\pi}\int_{\langle 2\pi\rangle} X(\Omega) e^{j\Omega n} d\Omega$

Ex: 0 2 5 $= x[n]$

$X(\Omega) = x[0]e^{-j\Omega 0} + x[1]e^{-j\Omega 1} + x[2]e^{-j\Omega 2} + \dots$

$= 2e^{-j\Omega} + 5e^{-j\Omega 2}$

$x[n] = \int_0^{2\pi}(2e^{-j\Omega} + 5e^{-j2\Omega}) e^{j\Omega n} d\Omega$

↓

just take n=7 $x[7] = \frac{1}{2\pi}\int_0^{2\pi}(2e^{-j\Omega} + 5e^{-j2\Omega}) e^{j7\Omega} d\Omega$

$= \frac{2}{2\pi}\int_0^{2\pi} e^{6j\Omega} d\Omega + \frac{5}{2\pi}\int_0^{2\pi} e^{5j\Omega} d\Omega$

$= \frac{2}{2\pi}\left.\frac{e^{6j\Omega}}{6j}\right|_0^{2\pi} + \frac{5}{2\pi}\left.\frac{e^{5j\Omega}}{5j}\right|_0^{2\pi}$

$= \frac{1}{\pi}\frac{1}{6j}\cancel{(e^{12\pi j} - e^{0})} + \frac{1}{2\pi j}\cancel{(e^{10\pi j} - e^{0})} = 0$

Ex: $x[n] = -4\ \underline{2}\ 3$ → [illegible] [illegible] sıfırda

$X(\Omega) = -4e^{j\Omega} + 2 + 3e^{-j\Omega}$

n=10 $x[10] = \frac{1}{2\pi}\int_0^{2\pi}(-4e^{j\Omega} + 2 + 3e^{-j\Omega}) e^{j10\Omega} d\Omega$

$= \frac{1}{2\pi}\int_0^{2\pi}(-4e^{j11\Omega} + 2e^{j10\Omega} + 3e^{-j9\Omega}) d\Omega$

✗

n=1 $x[1] = \frac{1}{2\pi}\int_0^{2\pi}(-4e^{-j\Omega} + 2 + 3e^{-j\Omega}) e^{j1\Omega} d\Omega$

$= 3$

Convolution Property:

$$\begin{tikzcd} x[n] \arrow[r] \arrow[d] & \boxed{h(n)} \arrow[r] \arrow[d] & x[n] * h[n] \arrow[d] \\ X(\Omega) \arrow[r] & \boxed{H(\Omega)} \arrow[r] & X(\Omega)\,H(\Omega)\ (\text{multiplied}) \end{tikzcd}$$

Example:

$x[n] = e^{-\alpha n} u[n]$ $\alpha > 0$ find its DFT

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

$$X(\Omega) = \sum_{n=0}^{\infty} e^{-\alpha n} e^{-j\Omega n} = \sum_{n=0}^{\infty} e^{-(\alpha + j\Omega)n}$$

$$= \frac{1}{1-\alpha-j\Omega}$$

Modulation Theory

Date: 09.06.2021

Amplitude Modulation (AM)

① $f(t) * \delta(t-t_0) = \underbrace{\int f(t-\tau)\,\delta(\tau-t_0)\,dt}_{\tau = t_0} = f(t-t_0)$

② $\boxed{\mathrm{IFT}[e^{jft}] \longrightarrow 2\pi\delta(\omega-f)}$

$$FT[e^{jft}] = \int_{-\infty}^{\infty} e^{jft} \cdot e^{-j\omega t}\,dt = 2 \cdot \lim_{T\to\infty} \frac{\sin((f-\omega)T)}{(f-\omega)}$$

$$\boxed{= 2\pi\delta(\omega-f)}$$

③ $FT(\cos(ft)) = \pi[\delta(\omega-f) + \delta(\omega+f)]$

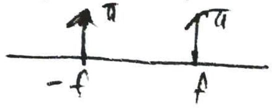
④ Human voice is limited to 20 kHz

time | **freq**

$s(t)\cdot\cos(ft)$ $\quad$ $\underbrace{FT(s(t)) * FT(\cos(ft))}$

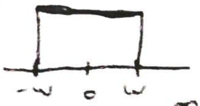
$\pi[\delta(\omega-f) + \delta(\omega+f)]$

$FT(x(t)) *$



gn ada karena 3 kali dengan gend, nd almadina.

Filters: Take the inverse FT of $X_{(-f,f)}(\omega)$



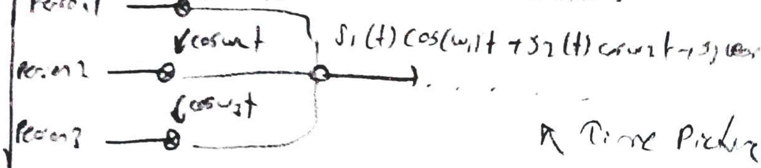
$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_{(-f,f)}(\omega)\, e^{j\omega t}\,d\omega = \frac{1}{2\pi}\int_{-f}^{f} e^{j\omega t}\,d\omega$$

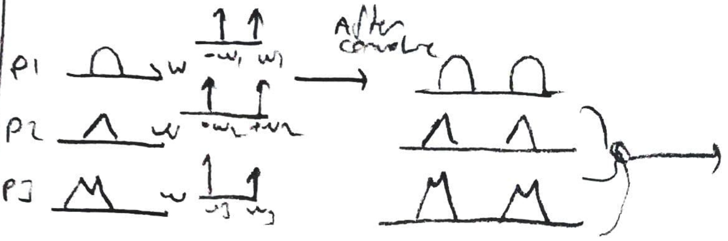
$$= \frac{1}{\pi}\frac{\sin(tf)}{t} \qquad FT\left[\frac{1}{\pi}\frac{\sin(tf)}{t}\right] = X_{(-f,f)}(\omega)$$

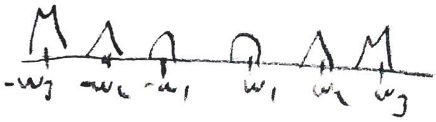
AMPLITUDE Modulation:

Date: 10.06.2021

modulation.


Person 1
Person 2
Person 3
cos w1t
cos w2t
cos w3t
S1(t) cos(w1t + S2(t) cos w2t + S3(t) cos
Time Pictur


P1
P2
P3
w
w
w
-w1 w1
-w2 +w2
-w3 w3
After convolve


-w3
-w2
-w1
w1
w2
w3